

The Bosonic Laughlin Spectral Gap: A Candidate Proof

An AI-generated research note

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Abstract

We present a candidate computer-assisted proof of $H_Q^2 \geq H_Q/3$ for the full spherical bosonic V_0 interaction at every integer flux $Q \geq 3088$, independently of particle number. Each unordered pair projector has coefficient one. At flux $Q = 2(N - 1)$ this gives a gap of at least $1/3$ above the unique bosonic Laughlin state. The argument combines 138 fixed operator rows, coherent comparisons resolved by total spin, and 156 parameter-interval block certificates. Six overlapping flux ranges cover $[3088, \infty)$. The certificate checks use exact rational arithmetic or outward-rounded integer intervals.

Introduction

The bosonic Laughlin state at filling $\nu = 1/2$ is an exact zero mode of the V_0 Haldane pseudopotential in the lowest Landau level. The wavefunction and spherical parent-Hamiltonian construction belong to the Laughlin–Haldane framework [1, 2]. The second-quantized pseudopotential algebra and its zero modes have also been studied directly [3, 4]. Here we consider the complete projected interaction on the round sphere, with coefficient one for each unordered pair projector onto relative angular momentum zero.

The spectral question is whether the positive spectrum stays uniformly separated from zero as the number of particles and the flux grow. Rougerie’s survey states the Laughlin spectral-gap conjecture and discusses its geometric variants [5, Appendix A]. For a truncated bosonic pseudopotential arising from the thin-cylinder regime, Warzel and Young proved gap bounds uniform in volume and particle number [6]. The present candidate proof concerns the full spherical V_0 interaction. Writing H_Q for this Hamiltonian at integer flux Q , its proposed estimate is $H_Q^2 \geq H_Q/3$ for $Q \geq 3088$, uniformly in the particle number N ; at $Q = 2(N - 1)$ this gives the stated gap above the Laughlin zero mode.

The argument adapts the comparison on the full Fock space in OpenAI’s preprint on the fermionic V_1 interaction [7]. Its structure is a normal-ordered Hamiltonian square, a rotation average of positive operator rows, finite spin-block comparisons, and transfer from planar coefficients to finite flux. For the bosonic model, coherent occupation bounds control the comparison errors before the many-body lift. The resulting finite inequalities are supplied as frozen data and checked with exact rational arithmetic and outward-rounded intervals.

***AI-generated candidate proof.** OpenAI Codex AI agents developed the core argument, verification code, threshold optimization, and manuscript under Xin Shen’s direction and curation on 7 October 2026. The session identifies the model family as GPT-6; the exact deployed model identifier was not recorded. Exact planar and scalar checks and 156 interval block certificates were replayed successfully. The analytical reduction and its implementation remain open to independent review; no independent human proof review or proof-assistant formalization is recorded for v0.1. See the repository’s AI contribution and verification statement.

1 Model, normalization, and three-body input

On the sphere, Q is the integer number of flux quanta. The one-particle LLL is U_Q , the spin- $Q/2$ representation, with normalized orbitals e_i , $0 \leq i \leq Q$. The physical N -particle space is $\text{Sym}^N U_Q$. The pair space $V_Q \subset \text{Sym}^2 U_Q$ is the maximal-spin- Q subspace, the subspace of relative angular momentum zero. Its normalized orbital-sum basis is v_p , $0 \leq p \leq 2Q$. With bosonic CCR $[a_i, a_j^\dagger] = \delta_{ij}$, set

$$t_p^Q(i, j) = \delta_{i+j, p} \sqrt{\frac{\binom{Q}{i} \binom{Q}{j}}{\binom{2Q}{p}}}, \quad B_p = \frac{1}{\sqrt{2}} \sum_{i,j=0}^Q t_p^Q(i, j) a_i a_j, \quad H_Q = \sum_p B_p^\dagger B_p. \quad (1)$$

Thus $B_p^\dagger |0\rangle = v_p$ and each unordered pair projector has coefficient one. The interaction is not orbitally truncated. At $Q = 2(N-1)$ its zero mode is the bosonic Laughlin polynomial $\prod_{i < j} (u_i v_j - u_j v_i)^2$, where (u_i, v_i) are homogeneous spinor coordinates of particle i on the sphere; an LLL polynomial is homogeneous of degree Q in each particle's spinor.

On the plane, U is the Bargmann LLL with normalized orbitals e_i , $i \geq 0$. Replace the tensor by

$$t_p^\infty(i, j) = \delta_{i+j, p} \sqrt{\frac{p!}{2^p i! j!}}, \quad H = \sum_{p \geq 0} B_p^\dagger B_p. \quad (2)$$

All forms below can first be read on finite-particle algebraic Fock vectors. In each fixed N sector the projected pair Hamiltonian is bounded. The particle-number-independent inequalities then hold on the appropriate direct-sum form domain.

For a normalized k -particle vector w , let $A(w)^\dagger$ be the unique homogeneous degree- k polynomial in creation operators satisfying $A(w)^\dagger |0\rangle = w$, and let $A(w)$ be its physical adjoint. The positive, normal-ordered k -body lift is the linear map

$$\mathcal{L}_k(|w\rangle\langle w'|) = A(w)^\dagger A(w'). \quad (3)$$

It takes positive operators on the physical k -particle space to positive forms on all bosonic particle sectors. In an occupation basis ν , $A(\nu) = \prod_i a_i^{\nu_i} / \sqrt{\nu_i!}$; no extra factorial is present.

Normal ordering gives the exact identity

$$H_Q^2 = H_Q + S_{3,Q} + S_{4,Q}, \quad S_{4,Q} = \sum_{p,q} (B_q B_p)^\dagger (B_q B_p) \geq 0. \quad (4)$$

The auxiliary space $V_Q \otimes U_Q$ contains a pair and a distinguished third particle. The linear creation map

$$W_3 : V_Q \otimes U_Q \longrightarrow \text{Sym}^3 U_Q, \quad v_p \otimes u \longmapsto B_p^\dagger u \quad (5)$$

is not an isometry. Couple pair spin Q and single-particle spin $Q/2$. Let Π_z be the auxiliary projection onto spin $J_z = 3Q/2 - z$, $0 \leq z \leq Q$, with dimension $d_z = 3Q - 2z + 1$. The recoupling identities are

$$W_3^\dagger W_3 = \sum_{z=0}^Q \lambda_{z,Q} \Pi_z, \quad \lambda_{z,Q} = 1 + 2(-1)^z f_{z,Q}, \quad f_{z,Q} = \frac{(Q)_z}{(2Q)_z}, \quad (6)$$

and

$$S_{3,Q} = \mathcal{L}_3 \left[W_3 \left(2 \sum_z (-1)^z f_{z,Q} \Pi_z \right) W_3^\dagger \right]. \quad (7)$$

Here $(x)_z = x(x-1) \cdots (x-z+1)$, with $(x)_0 = 1$. Let $J : V_Q \otimes U_Q \rightarrow U_Q^{\otimes 3}$ be the inclusion putting the pair in the first two tensor slots, and let P_{sym} be the orthogonal three-particle symmetrizer.

Then $W_3 = \sqrt{3}P_{\text{sym}}J$ and $W_3^\dagger W_3 = I + 2T$, where $T = J^\dagger(23)J$ is the compressed transposition of slots two and three. The pair–single spin decomposition is multiplicity free. A highest vector at deficit z has the form $\sum_{p=0}^z s_p v_p \otimes e_{z-p}$, with

$$\frac{s_{p+1}}{s_p} = -\sqrt{\frac{(z-p)(Q-z+p+1)}{(p+1)(2Q-p)}}, \quad \frac{s_z}{s_0} = (-1)^z \sqrt{f_{z,Q}}. \quad (8)$$

Taking the $v_0 \otimes e_z$ coefficient after transposition gives $s_z t_z^Q(0, z) = s_z \sqrt{f_{z,Q}}$; hence T has eigenvalue $(-1)^z f_{z,Q}$. This proves (6). Also $W_3 W_3^\dagger = H_{3,Q}$, so its three-body square minus itself is the displayed three-body coefficient. Direct CCR normal ordering gives $[B_p, B_q^\dagger] = \delta_{pq} + 2 \sum_{i,j,k} t_p^Q(i, k) t_q^Q(j, k) a_j^\dagger a_i$, and consequently (4) on every particle sector by the positive three-body lift. In particular, S_3 is not positive; the $z = 1$ auxiliary block is annihilated by W_3 .

Fix an odd integer $m \geq 3$. The positive magnitude of the negative tail is

$$C_{m,Q} = 2 \sum_{\substack{z \geq m \\ z \text{ odd}}} f_{z,Q} \Pi_z, \quad T_{m,Q} = \mathcal{L}_3(W_3 C_{m,Q} W_3^\dagger). \quad (9)$$

In the plane, $f_{z,\infty} = 2^{-z}$. The index z there is the degree of the internal oscillator between pair center of mass and the third particle.

2 Plane estimate

Theorem 1. *For odd $m \geq 3$ and every particle number,*

$$T_m \leq c_m(S_4 + H), \quad c_m = 3(m+1) \left(\frac{m+1}{2m} \right)^m. \quad (10)$$

More generally, for $1/2 < r < 1$ and $\eta > 0$,

$$T_m \leq \frac{3}{1-r} \left(\frac{1}{2r} \right)^m (\eta S_4 + C_\eta H), \quad C_\eta = \max_{n \in \mathbb{Z}_{\geq 0}} \left[n - \frac{\eta}{2} n(n-1) \right]. \quad (11)$$

In particular $C_1 = 1$.

Proof. Use the pair center-of-mass oscillator u and third-particle oscillator v . The unitary oscillator change is

$$\zeta = \frac{\sqrt{2}u + v}{\sqrt{3}}, \quad \xi = \frac{-u + \sqrt{2}v}{\sqrt{3}}. \quad (12)$$

Let n_R count quanta of ξ . The auxiliary tail satisfies

$$C_m = 2 \cdot 2^{-n_R} P_{\text{odd}, \geq m} \leq 2 \left(\frac{1}{2r} \right)^m (I_{\text{CM}} \otimes r^{n_R}). \quad (13)$$

For normalized oscillator coherent vectors the exact positive representation is

$$r^{n_R} = \int \frac{d^2 \xi}{\pi r} e^{-(1-r)|\xi|^2/r} |\xi\rangle \langle \xi|. \quad (14)$$

Its j -th diagonal matrix element is r^j , obtained by the elementary Gaussian moment integral. Resolve I_{CM} in coherent states as well. Write $u = \sqrt{2}\alpha$, $v = \beta$, where α, β are physical one-particle coherent coordinates. Then

$$I_{\text{CM}} \otimes r^{n_R} = \int \frac{2d^2 \alpha d^2 \beta}{\pi^2 r} e^{-\kappa|\beta-\alpha|^2} |\sqrt{2}\alpha\rangle_V \langle \sqrt{2}\alpha| \otimes |\beta\rangle \langle \beta|, \quad \kappa = \frac{2(1-r)}{3r}. \quad (15)$$

The factor 2 is the Jacobian of $u = \sqrt{2}\alpha$.

Let a_α annihilate in the normalized physical coherent orbital, $B_\alpha = a_\alpha^2/\sqrt{2}$, and $n_\beta = a_\beta^\dagger a_\beta$. The pair coherent vector is exactly $B_\alpha^\dagger|0\rangle$. Positive lifting the integrand in (15) after W_3 produces $B_\alpha^\dagger n_\beta B_\alpha$. Since the coherent pair belongs to V ,

$$H \geq B_\beta^\dagger B_\beta = \frac{1}{2}n_\beta(n_\beta - 1), \quad n_\beta \leq \eta H + C_\eta I. \quad (16)$$

The latter inequality acts on the output sector after B_α removes two particles. The Gaussian β marginal is $3/[2(1-r)]$. The pair coherent resolution gives the exact identities

$$\int \frac{2d^2\alpha}{\pi} B_\alpha^\dagger B_\alpha = H, \quad \int \frac{2d^2\alpha}{\pi} B_\alpha^\dagger H B_\alpha = \sum_p B_p^\dagger H B_p = S_4. \quad (17)$$

Pair annihilators commute, which explains the last equality. Consequently

$$\mathcal{L}_3[W_3(I_{\text{CM}} \otimes r^{n_R})W_3^\dagger] \leq \frac{3}{2(1-r)}(\eta S_4 + C_\eta H). \quad (18)$$

Combine with the auxiliary spectral bound. At $\eta = 1$, minimize over r ; the minimum occurs at $r = m/(m+1)$ and gives (10). $\square$

3 Spherical estimate with the same asymptotic constant

Theorem 2. Fix odd $m \geq 3$ and $\varepsilon > 0$. There exists a flux threshold $Q_{m,\varepsilon}$ independent of particle number such that

$$T_{m,Q} \leq (c_m + \varepsilon)(S_{4,Q} + H_Q) \quad (Q \geq Q_{m,\varepsilon}) \quad (19)$$

on all bosonic particle sectors, with the same c_m as in Theorem 1. The threshold is existential in this argument.

Proof. Let $d = Q + 1$, $d_a = 2Q + 1$. Sphere measures have total mass one. Fix $0 < a < 1/3$, $M = \lfloor aQ \rfloor$, and introduce the positive auxiliary kernel

$$A_{M,Q} = d_a d \int d\Omega_\alpha d\Omega_\beta s(\alpha, \beta)^M |\alpha_{\text{pair}}\rangle \langle \alpha_{\text{pair}}| \otimes |\beta\rangle \langle \beta|, \quad s = \cos^2(\theta_{\alpha\beta}/2). \quad (20)$$

These are normalized spin-coherent vectors. Rotation invariance gives $A_{M,Q} = \sum_z \tau_{z,Q} \Pi_z$.

Fix α at the north pole. A beta integral gives the diagonal coefficient

$$b_{j,Q} = d \binom{Q}{j} \int_0^1 s^{M+Q-j} (1-s)^j ds = \frac{Q+1}{M+Q+1} \frac{(Q)_j}{(M+Q)_j}. \quad (21)$$

Tracing in the coupled spin block yields

$$\tau_{z,Q} = \frac{d_a}{d_z} \sum_{j=z}^Q b_{j,Q} \gamma_{z,j,Q}, \quad \gamma_{z,j,Q} = \frac{(2Q)_z}{(3Q-z+1)_z} \binom{j}{z} \frac{(Q-z)_{j-z}}{(3Q-2z)_{j-z}}. \quad (22)$$

Here γ is the squared Clebsch–Gordan coefficient for a highest pair and a single orbital lowered j times. To derive it, normalize the highest coupled-spin vector using

$$\frac{|c_p|^2}{|c_{p-1}|^2} = \frac{(z-p+1)(Q-z+p)}{p(2Q-p+1)}. \quad (23)$$

The terminating Chu–Vandermonde identity gives $|c_0|^2 = (2Q)_z/(3Q-z+1)_z$. After total lowering $j-z$ times, only lowering the single-particle component can retain pair highest weight; the normalization ratio gives the remaining binomial and falling-factorial factors in (22).

We first need an all- z coarse bound. Retaining only the positive $j = z$ term in (22), with $\rho = 3(1+a)/4 < 1$, gives

$$\frac{2f_{z,Q}}{\tau_{z,Q}} \leq 3(1+a)\rho^z. \quad (24)$$

Indeed the ratio before the factor 2 is at most

$$\frac{d_z}{d_a} \frac{M+Q+1}{d} \prod_{i=0}^{z-1} \frac{(M+Q-i)(3Q-z+1-i)}{(2Q-i)^2}. \quad (25)$$

The prefactor is at most $\frac{3}{2}(1+a)$. Since $i \leq z-1$ and $M \leq aQ$, each product factor is at most $(1+a-x)(3-2x)/(2-x)^2 \leq \rho$, $x = i/Q \in [0, 1]$. The last inequality is equivalent to $x[(8-4a)-(5-3a)x] \geq 0$.

For each fixed z , the full sum in (22) has the limit

$$\tau_{z,Q} \longrightarrow r^{z+1}, \quad r = \frac{2}{2+3a}. \quad (26)$$

For fixed z, j , the limiting summand uses $b_{j,Q} \rightarrow (1+a)^{-j-1}$ and $\gamma_{z,j,Q} \rightarrow \binom{j}{z}(2/3)^z(1/3)^{j-z}$, while $d_a/d_z \rightarrow 2/3$. The binomial-series identity sums these limits. To justify exchanging limit and sum, extend summands by zero for $j > Q$. For $Q \geq 2/a$, $M \geq aQ/2$,

$$0 \leq b_{j,Q} \leq (1+a/2)^{-j}, \quad 0 \leq \gamma_{z,j,Q} \leq 1, \quad d_a/d_z < 2. \quad (27)$$

This provides a summable majorant independent of Q .

Now fix m and choose $a = 2/(3m)$, so $r = m/(m+1) > 1/2$. The fixed- z ratio limit is $(2/r)(1/(2r))^z$, decreasing on odd $z \geq m$. Pick a finite L so large that the right side of (24) for $z > L$ is at most its limiting maximum at $z = m$. On the finite set $m \leq z \leq L$ use (26). For any $\delta > 0$, sufficiently large Q therefore gives the whole-space inequality

$$C_{m,Q} \leq \left[\frac{2}{r} \left(\frac{1}{2r} \right)^m + \delta \right] A_{M,Q}. \quad (28)$$

Finally lift the coherent kernel. Equation (16) holds on the sphere as well. The exact beta marginal is $d/(M+1)$, while the pair coherent resolution has factor d_a . Thus

$$\mathcal{L}_3(W_3 A_{M,Q} W_3^\dagger) \leq \frac{d}{M+1} (S_{4,Q} + H_Q). \quad (29)$$

Since $d/(M+1) \rightarrow 1/a$, the resulting scalar tends to $(2/(ra))(1/(2r))^m = c_m$. Choose δ small and then Q sufficiently large. All comparisons precede lifting and their constants are independent of N , proving (19). $\square$

4 Absorbing finite three- and four-body comparison errors

Write $H_{r,Q}$ for H_Q restricted to the physical r -particle space, where $r = 3, 4$. Let $\Pi_{\leq D,Q}^{(r)}$ be its total-spin projection onto $J = rQ/2 - z$, $0 \leq z \leq D$, including all multiplicities. Define the many-body positive forms

$$\mathcal{R}_{r,D,Q} := \mathcal{L}_r(H_{r,Q} \Pi_{\leq D,Q}^{(r)}). \quad (30)$$

The product is positive because the Hamiltonian is rotationally invariant. The cutoff is in the few-body coefficient, not in the physical many-body Hamiltonian or in particle occupation.

Lemma 1. *For all sufficiently large Q , uniformly in total particle number,*

$$\mathcal{R}_{3,8,Q} \leq 35(S_{4,Q} + H_Q), \quad \mathcal{R}_{4,16,Q} \leq 2100(2S_{4,Q} + H_Q). \quad (31)$$

Both estimates hold, in particular, for every integer $Q \geq 10^{50}$, as shown in Appendix A.2. Their thresholds depend only on the displayed fixed cutoffs.

Proof. The three-body claim follows from the positive auxiliary operator $A_{M,Q}$ of the preceding section. On each retained spin block its eigenvalue tends to r^{z+1} , $r = 2/(2+3a)$. Since only $z \leq D$ is now retained, positivity outside this finite set and finite-block convergence give

$$\Pi_{\leq D,Q}^{\text{aux}} \leq (r^{-(D+1)} + o(1))A_{M,Q}. \quad (32)$$

After positive lifting, the limiting constant is $a^{-1}(1+3a/2)^{D+1}$. Taking $D=8, a=1/12$ gives $12(9/8)^9 = 34.63809 \dots < 35$.

For the four-body claim, use the wide pair map

$$W_4 : V_Q \otimes \text{Sym}^2 U_Q \longrightarrow \text{Sym}^4 U_Q, \quad W_4(v_p \otimes u) = B_p^\dagger u. \quad (33)$$

It satisfies $W_4 W_4^\dagger = H_{4,Q}$. Its domain differs from the narrower pair-pair space that represents S_4 . Equivariance gives $W_4 \Pi_{\leq D,Q}^{\text{aux}} W_4^\dagger = H_{4,Q} \Pi_{\leq D,Q}^{(4)}$.

First establish an occupation bound for possibly nonorthogonal coherent orbitals β, γ . Put $A = a_\gamma a_\beta$ and $s = \langle \beta, \gamma \rangle$, with $|s| \leq 1$. Normal ordering and $(n_\beta - n_\gamma)^2 \geq 0$ imply

$$\begin{aligned} A^\dagger A &= \frac{1}{2}(n_\beta n_\gamma + n_\gamma n_\beta) - \text{Re}(s a_\beta^\dagger a_\gamma) \\ &\leq \frac{1}{2}(n_\beta^2 + n_\gamma^2) + \frac{1}{2}(n_\beta + n_\gamma) \leq 4H_Q + 2I. \end{aligned} \quad (34)$$

For the middle inequality, quadratic-form Cauchy-Schwarz bounds the real cross term by $|s|(n_\beta + n_\gamma)/2$. The last step uses $n_x^2 \leq 2H_Q + n_x$ and $n_x \leq H_Q + I$ for each coherent x .

Set $M = \lfloor aQ \rfloor$ and introduce the positive invariant operator on $V_Q \otimes \text{Sym}^2 U_Q$

$$\begin{aligned} A_{M,Q}^{(4)} &= d_a d^2 \int d\Omega_\alpha d\Omega_\beta d\Omega_\gamma [s(\alpha, \beta)s(\alpha, \gamma)]^M |\alpha_{\text{pair}}\rangle \langle \alpha_{\text{pair}}| \\ &\quad \otimes P_{\text{sym}}^{(2)} |\beta\gamma\rangle \langle \beta\gamma| P_{\text{sym}}^{(2)}. \end{aligned} \quad (35)$$

The isometry from symmetric tensors to bosonic Fock space sends $P_{\text{sym}}^{(2)} |\beta\gamma\rangle$ to $a_\beta^\dagger a_\gamma^\dagger |0\rangle / \sqrt{2}$. Consequently the lifted coherent integrand is $\frac{1}{2} B_\alpha^\dagger A^\dagger A B_\alpha$, and (34), the exact two spectator marginals, and the pair frame give

$$\mathcal{L}_4(W_4 A_{M,Q}^{(4)} W_4^\dagger) \leq \left(\frac{d}{M+1} \right)^2 (2S_{4,Q} + H_Q). \quad (36)$$

We next compute the limit of every fixed auxiliary spin block, including its multiplicities. Choose continuous normalized highest-weight bases at deficit z , and normalized total-lowering descendants numbered by n . Such bases exist for large Q : the finite-weight raising matrices converge to the oscillator lowering matrix, which has full row rank. More explicitly, the auxiliary weight- z basis consists of $v_p \otimes |j, k\rangle_{\text{sym}}$, $p+j+k=z$, $j \leq k$. For $Q \geq z$ its dimension is fixed, and the highest-weight multiplicity is $\lfloor z/2 \rfloor + 1$. After division by $\sqrt{Q}$, the raising matrix tends to $2a_\zeta$, where a_ζ annihilates the total center-of-mass oscillator introduced below. It is surjective from weight z to weight $z-1$. For a full-row-rank matrix T , the kernel projector is $I - T^\dagger (T T^\dagger)^{-1} T$; this formula proves convergence of these kernel projections and permits convergent orthonormal bases. The case $z=0$ is the one-dimensional highest line and needs no inverse. Normalized spherical descendants then converge, for each fixed descendant number n , to $(a_\zeta^\dagger)^n / \sqrt{n!}$ applied to the limiting highest vectors. Fix the pair at the north pole. The two-spectator operator is the restriction of $R_Q \otimes R_Q$ to symmetric tensors, where R_Q is diagonal with eigenvalues $b_{j,Q}$ from the preceding section. Schur averaging expresses each multiplicity-matrix entry as

$$\frac{d_a}{4Q - 2z + 1} \sum_{n \geq 0} \sum_{\substack{j \leq k \\ j+k=z+n}} b_{j,Q} b_{k,Q} \overline{c_{b,jk,n,Q}} c_{c,jk,n,Q}. \quad (37)$$

Here $c_{b,jk,n,Q}$ is the pair-highest component of the normalized descendant of highest vector b ; the spectator basis is orthonormal, also when $j = k$. The squared sum of these components is at most one. For $Q \geq 2/a$, the inner sum is therefore bounded in absolute value by q^{z+n} , where $q = (1 + a/2)^{-1} < 1$. The dimension ratio is bounded. This summable majorant justifies dominated convergence of the entire descendant sum.

To identify that limit, use the unitary Jacobi-oscillator change

$$u = \sqrt{2}\alpha, \quad v = \beta, \quad w = \gamma, \quad \zeta = \frac{\sqrt{2}u + v + w}{2}, \quad \xi_1 = \frac{v - w}{\sqrt{2}}, \quad \xi_2 = \frac{u}{\sqrt{2}} - \frac{v + w}{2}. \quad (38)$$

In this limit the pair-north seed is the positive trace-class operator

$$\sigma = |0\rangle_u \langle 0| \otimes [R \otimes R]_{\text{sym}}, \quad R = (1 + a)^{-(n+1)} = \int \frac{d^2\beta}{\pi} e^{-a|\beta|^2} |\beta\rangle \langle \beta|, \quad (39)$$

where n is the one-oscillator occupation operator in either spectator factor. On relative degree z , the full descendant sum is the restriction of the center-of-mass partial trace of σ , with the prefactor $d_a/(4Q - 2z + 1) \rightarrow 1/2$. Let $D_\zeta(h) = \exp(ha_\zeta^\dagger - \bar{h}a_\zeta)$ be the Weyl displacement on the center-of-mass factor. Simultaneously displacing (u, v, w) by $(\sqrt{2}\alpha, \alpha, \alpha)$ displaces only ζ , by 2α . The Weyl twirling identity for the positive trace-class seed therefore gives, as a weak operator integral,

$$\int \frac{2d^2\alpha}{\pi} D_\zeta(2\alpha) \sigma D_\zeta(2\alpha)^\dagger = \frac{1}{2} I_{\text{CM}} \otimes \text{Tr}_{\text{CM}} \sigma. \quad (40)$$

The factor $1/2$ follows from $d^2(2\alpha) = 4d^2\alpha$ and agrees with the Schur dimension ratio. Inserting the coherent representation of R into the left side of (40) gives the measure $2d^2\alpha d^2\beta d^2\gamma/\pi^3$ with weight $e^{-a(|\beta-\alpha|^2+|\gamma-\alpha|^2)}$. The twirling identity can also be checked in center-of-mass position coordinates x, y . For $h = (q + ip)/\sqrt{2}$, conjugation by $D_\zeta(h)$ sends the operator-valued kernel to $e^{ip(x-y)} \sigma(x-q, y-q)$. Integrating $dq dp/(2\pi)$ first gives $\delta(x-y)$ and then the partial trace. Here the Gaussian, positive trace-class seed permits this identity in the weak sense. The substitution $u = \sqrt{2}\alpha$ cancels this factor two. Thus the limiting coherent kernel has weight $\exp[-a|\xi_1|^2 - 2a|\xi_2|^2]$, and hence is exactly

$$A_\infty^{(4)} = I_{\text{CM}} \otimes r_1^{n_1+1} r_2^{n_2+1}, \quad r_1 = (1 + a)^{-1}, \quad r_2 = (1 + 2a)^{-1}. \quad (41)$$

Here $n_i = a_{\xi_i}^\dagger a_{\xi_i}$ counts quanta of relative oscillator ξ_i , for $i = 1, 2$. Spectator symmetry restricts n_1 to even values. At fixed deficit $n_1 + n_2 = z$, the least eigenvalue is $r_1 r_2^{z+1}$. Thus on the finite set $z \leq D$, continuity gives

$$\Pi_{\leq D, Q}^{\text{aux}} \leq [(r_1 r_2^{D+1})^{-1} + o(1)] A_{M, Q}^{(4)}. \quad (42)$$

The inequality holds on the whole auxiliary space because invariance prohibits cross terms between distinct spins and the right side is positive outside the retained blocks. Combining with (36), the limiting scalar is

$$C_{4, D}(a) = \frac{(1 + a)(1 + 2a)^{D+1}}{a^2}. \quad (43)$$

At $D = 16, a = 1/16$, this equals $272(9/8)^{17} = 2014.47455 \dots < 2100$. All constants and thresholds were obtained before the many-body lift and are independent of total particle number. $\square$

5 Frozen finite certificate and its exact verification

We use rows with net orbital weight $t \in \{0, 1, 2, 3\}$,

$$F_\rho = \lambda_\rho B_{t_\rho} + \sum_{\substack{p, j \geq 0 \\ t_\rho \leq p+j \leq 8}} \alpha_{pj}^\rho a_{p+j-t_\rho}^\dagger a_j B_p. \quad (44)$$

All coefficients are fixed real algebraic numbers, independent of Q, N . They are defined exactly by the companion file `exact_shifted_certificate_rows.json`, as specified below. The four channels contain 31, 37, 36, 34 rows, respectively, totaling 138. Set

$$K_Q = (2Q + 1) \int \sum_{\rho} F_{\rho}(g)^{\dagger} F_{\rho}(g) dg = \eta H_Q + K_{3,Q} + K_{4,Q} \geq 0, \quad \eta = \sum_{\rho} \lambda_{\rho}^2. \quad (45)$$

Haar measure has mass one. Normal ordering terminates at body order four. Here $g \in SU(2)$ and $F_{\rho}(g) = U(g)F_{\rho}U(g)^{\dagger}$, with $U(g)$ the rotation representation on the direct sum of all particle sectors. Before averaging, the three-body coefficient has orbital-weight support at most 8, and the four-body coefficient at most 16. After $SU(2)$ averaging their spin-deficit supports are $z \leq 8$ and $z \leq 16$, because an irrep of deficit z contains no orbital-weight space $M < z$. The averaged comparison coefficients extend over all descendants of the retained irreps; H_Q remains untruncated. Suppress the row index and put $i = p + j - t$, $i' = p' + j' - t$. For a real row, its unaveraged normal-ordered terms beyond $\lambda^2 B_t^{\dagger} B_t$ are

$$\begin{aligned} K_3^{\text{row}} &= \lambda \sum_{p,j} \alpha_{pj} (B_t^{\dagger} a_i^{\dagger} a_j B_p + \text{h.c.}) \\ &\quad + \sum_{p,j,p',j'} \alpha_{pj} \alpha_{p'j'} \delta_{ii'} B_p^{\dagger} a_j^{\dagger} a_{j'} B_{p'}, \\ K_4^{\text{row}} &= \sum_{p,j,p',j'} \alpha_{pj} \alpha_{p'j'} B_p^{\dagger} a_j^{\dagger} a_{i'}^{\dagger} a_i a_{j'} B_{p'}. \end{aligned} \quad (46)$$

The three-body weights are $p + j \leq 8$ and the four-body weights are $(p + j) + (p' + j') - t \leq 16$. Every term retains a pair annihilator on the right and a pair creator on the left. These are exact CCR identities on each sphere.

Let $K_{r,Q}$ denote the physical r -particle coefficient whose lift is $K_{r,Q}$. Let $S_{3,Q} = H_{3,Q}^2 - H_{3,Q}$ and let $S_{4,Q}$ be the physical four-body coefficient of $S_{4,Q}$. Define

$$\begin{aligned} E_{3,Q} &= \Pi_{\leq 8,Q}^{(3)} S_{3,Q} \Pi_{\leq 8,Q}^{(3)} - K_{3,Q}, \\ E_{4,Q} &= \Pi_{\leq 16,Q}^{(4)} [(1 - c_9 - \mu) S_{4,Q} - K_{4,Q}] \Pi_{\leq 16,Q}^{(4)}, \end{aligned} \quad (47)$$

$$c_9 = 30(5/9)^9 = \frac{19531250}{129140163}, \quad \mu = \frac{1}{50}. \quad (48)$$

Outside the four-body cutoff the comparison coefficient is a positive multiple of $S_{4,Q}$.

The finite certificate is evaluated in planar limiting relative blocks. If $L = \sum_{j \geq 1} \sqrt{j} a_{j-1}^{\dagger} a_j$ on the r -particle space, then $[L, L^{\dagger}] = r$. A highest (relative) vector at degree z has $L\psi = 0$, and its normalized degree- M descendant is $(L^{\dagger})^{M-z} \psi / \sqrt{r^{M-z} (M-z)!}$. Denote this isometric map from the relative highest-weight space to the degree- M physical space by $U_{r,z}^{(M)}$. In the next formula, $K_{r,M}$ is the unaveraged normal-ordered body coefficient restricted to total orbital degree M ; the overbar denotes its averaged relative block. Schur averaging on the sphere, followed by the fixed-block limit, gives

$$\bar{K}_{r,z} = \frac{2}{r} \sum_{M \geq z} (U_{r,z}^{(M)})^{\dagger} K_{r,M} U_{r,z}^{(M)}. \quad (49)$$

The sum is finite. The factor is $(2Q+1)/(rQ-2z+1) \rightarrow 2/r$. This also follows from the oscillator center-of-mass partial trace. Two-body averaging is exactly ηH .

To make the verification entirely rational, introduce canonical polynomial operators

$$x_j = \sqrt{j!} a_j^{\dagger}, \quad \partial_j = a_j / \sqrt{j!}. \quad (50)$$

They satisfy $[\partial_i, x_j] = \delta_{ij}$, but are not physical adjoints: $\partial_j^* = x_j/j!$ and $x_j^* = j! \partial_j$. For the unnormalized physical occupation basis $|A]_x = \prod_j x_j^{n_j} |0\rangle$, the metric is the positive integer $g_A =$

$\prod_j n_j!(j!)^{n_j}$. On the plane,

$$B_p = \kappa_p \hat{B}_p, \quad \kappa_p^2 = \frac{p!}{2^{p+1}}, \quad \hat{B}_p = \sum_{i+j=p} \partial_i \partial_j. \quad (51)$$

The frozen JSON integers divided by 10^{12} define the scaled row coefficients

$$y_0 = \lambda \kappa_t, \quad y_{pj} = \alpha_{pj} \kappa_p \sqrt{j!/i!}, \quad i = p + j - t. \quad (52)$$

Equations (44) and (52) uniquely define the same fixed coefficients used on every sphere.

For the finite verification, compute F^*F in the 2,3,4-particle sectors with the metric just given. Here F^*F is summed over all rows, and $\mathcal{L}_r^N$ denotes $\mathcal{L}_r$ restricted to the physical N -particle sector. Extract the normal-ordered coefficients recursively:

$$\begin{aligned} \mathsf{K}_2 &= (F^*F)|_{N=2}, \\ \mathsf{K}_3 &= (F^*F)|_{N=3} - \mathcal{L}_2^3(\mathsf{K}_2), \\ \mathsf{K}_4 &= (F^*F)|_{N=4} - \mathcal{L}_2^4(\mathsf{K}_2) - \mathcal{L}_3^4(\mathsf{K}_3). \end{aligned} \quad (53)$$

The coefficient at body order r is recovered because its normalized lift restricts to itself on r particles. There are no higher body orders in F^*F . The integer CM operators are $L = \sum_{j \geq 1} j x_{j-1} \partial_j$ and $L^\dagger = \sum_{j \geq 0} x_{j+1} \partial_j$. Rational row reduction gives a highest-weight basis. To use (49) without square roots, use unnormalized descendants and divide their quadratic forms by $r^{M-z}(M-z)!$.

With $\delta = 10^{-6}$, the exact rational tests give

$$E_{3,\infty} + \delta H_{3,\infty} \Pi_{\leq 8,\infty}^{(3)} \geq 0, \quad E_{4,\infty} + \delta H_{4,\infty} \Pi_{\leq 16,\infty}^{(4)} \geq 0. \quad (54)$$

All $9 + 17 = 26$ relative blocks were tested, including zero-dimensional ones. Positive-pivot rational Schur elimination establishes semidefiniteness; no floating eigenvalue tolerance enters. In every block the rank equals the rank of H_r , and the exact kernel of H_r is preserved. Thus each shifted matrix is strictly positive on the physical image of H_r . The exact two-body cost is

$$\eta = \frac{764191703750823432978457}{1500000000000000000000000000} < \frac{51}{100}. \quad (55)$$

The full rational pivots, ranks, and zero-kernel checks are in `exact_shifted_certificate_results.json`.

Only three adjacent files are needed to replay the proof of (54): the frozen JSON rows, `exact_relative_blocks.py`, and `exact_shifted_certificate.py`. Running

```
python -S exact_shifted_certificate.py
```

uses Python integers and `fractions.Fraction` to verify the frozen rows. The optional proposal mode is separate from the default verifier. The verifier returns a nonzero exit status on failure.

6 Transferring the finite certificate to spheres

Fix $r = 3, 4$ and $z \leq 16$. In the sphere weight- z occupation basis, all pair coefficients and the rescaled raising/lowering matrices converge to their planar counterparts. For $Q \geq z$ the finite weight spaces have fixed dimensions. Dividing the spherical lowering operator by $\sqrt{Q}$ gives $L^\dagger$ in the limit, and normalizing descendants gives (49). The finite normal-order supports make this a finite sum for the certificate.

The relevant zero spaces also have exactly stable dimensions. In the bosonic LLL, $H_{r,Q}\psi = 0$ if and only if the polynomial vanishes at every pair coincidence. Symmetry makes this vanishing at least quadratic, so the polynomial is divisible by

$$\Delta_r^2 = \prod_{i < j} (u_i v_j - u_j v_i)^2. \quad (56)$$

Conversely this divisibility annihilates every V_0 pair channel. The factor is an $SU(2)$ invariant and reduces each single-particle degree by $2(r-1)$. Let $p_r(z)$ count partitions of z into at most r parts, with $p_r(z) = 0$ for $z < 0$, and put $d_r(z) = p_r(z) - p_r(z-1)$. For $Q \geq z$, the relative physical dimension is $d_r(z)$ and the relative zero-space dimension is $d_r(z - r(r-1))$. Thus the active rank is exactly

$$d_r(z) - d_r(z - r(r-1)), \quad (57)$$

the same as on the plane. Indeed multiplication by Δ_r^2 is an injective $SU(2)$ intertwiner from the residual symmetric polynomial space with single-particle degree $Q' = Q - 2(r-1)$. It shifts total orbital degree by $r(r-1)$, while preserving total spin. If $z \geq r(r-1)$ and $Q \geq z$, the residual weight $z' = z - r(r-1)$ obeys $z' \leq Q'$, so the orbital cap is inactive in both weight spaces used to compute the highest-weight multiplicity. If $z < r(r-1)$ there are no zero modes in this block.

The unprojected normal-ordered coefficients defining $E_{3,Q}, E_{4,Q}$ have a pair annihilator on the right and a pair creator on the left. The spin projections preserve $\ker H_{r,Q}$ because $H_{r,Q}$ is rotationally invariant. Thus $E_{r,Q}$ annihilates this kernel exactly. The kernel may rotate with Q , but the ranks in (57) are constant. Identify the highest spaces by convergent orthonormal frames. In one block write H_Q, E_Q for the corresponding matrices, P_Q for the orthogonal projection onto $\text{ran } H_Q$, and $G_Q = E_Q + \delta H_Q$ with the planar allowance $\delta = 10^{-6}$. Stable rank and norm convergence imply $P_Q \rightarrow P_\infty$: the positive spectrum of H_∞ is separated from zero, and its spectral cluster projection is continuous; rank equality rules out additional small positive eigenvalues. Exact kernel preservation gives $G_Q = P_Q G_Q P_Q$. On any nonzero active block let $g > 0$ be the least eigenvalue of G_∞ restricted to $\text{ran } H_\infty$, whose strict positivity was verified exactly. For a unit vector $x \in \text{ran } H_Q$,

$$\langle x, G_Q x \rangle \geq g(1 - \|P_Q - P_\infty\|^2) - \|G_Q - G_\infty\|. \quad (58)$$

This follows from $G_\infty \geq gP_\infty$ and $\|(I - P_\infty)x\| \leq \|P_Q - P_\infty\|$. For all sufficiently large Q , both norms are small enough to make the right side positive. A zero-rank block is identically zero by kernel preservation. Thus $G_Q \geq 0$ on the entire block, including its rotating zero space. In particular the following slightly weaker bounds hold with a fixed allowance $\delta_Q = 2 \times 10^{-6}$:

$$E_{3,Q} \geq -\delta_Q H_{3,Q} \Pi_{\leq 8,Q}^{(3)}, \quad E_{4,Q} \geq -\delta_Q H_{4,Q} \Pi_{\leq 16,Q}^{(4)}. \quad (59)$$

There are only finitely many blocks. The threshold depends on their matrices and the fixed certificate and is uniform in the total particle number to which these inequalities are lifted. Appendix A.3 gives a quantitative proof using exact polynomial coordinates and an additional finite planar half-gap check; it establishes these same inequalities for every integer $Q \geq 10^{50}$.

7 The resulting Fock-space inequality

Theorem 3. *For the full spherical bosonic V_0 interaction with coefficient one per unordered pair projector, the explicit integer threshold $Q_0 = 3088$ works independently of particle number:*

$$H_Q^2 \geq \frac{1}{3} H_Q \quad (Q \geq Q_0) \quad (60)$$

on the entire bosonic LLL Fock space, as a quadratic-form inequality. For $Q = 2(N-1) \geq Q_0$,

$$H_{N,Q} \geq \frac{1}{3} (I - P_{L,N}), \quad (61)$$

where $P_{L,N}$ projects onto the normalized bosonic Laughlin state. The plane interaction also satisfies $H^2 \geq H/3$ on every particle sector.

Proof. For $Q \geq 3088$, select a row of Table 1 whose flux interval contains Q . Write L_3, L_4 for its two exact rational error budgets and g for its tail improvement. The row ending at a finite flux q_+ uses $g = 391/(100q_+)$; the last row uses $g = 0$. Put $\nu = 10^{-10}$ and retain $\mu = 1/50$.

Appendix B supplies a direct certificate on the entire parameter interval $0 \leq 1/Q \leq 1/q_-$ for each row, where q_- is its lower endpoint. Its comparison error in spin block z is the exact rational allowance $\delta_{r,z} = L_r/\overline{C}_{r,z}(q_-)$. The degree-resolved coherent comparison in Appendix B.2 absorbs these errors with cost $L_3(S_{4,Q} + H_Q)$ and $L_4(2S_{4,Q} + H_Q)$, respectively. The common positive coherent kernel dominates all its orthogonal spin blocks, so the cost is the maximum over z . Outside the retained blocks, the only negative contribution is the three-body tail. Thus the positive lifts obey

$$\begin{aligned} S_{3,Q} - K_{3,Q} &\geq -L_3(S_{4,Q} + H_Q) - T_{9,Q}, \\ (1 - c_9 - \mu)S_{4,Q} - K_{4,Q} &\geq -L_4(2S_{4,Q} + H_Q). \end{aligned} \quad (62)$$

The exact tail improvement, valid for all $Q \geq 3000$, is

$$T_{9,Q} \leq (c_9 + \nu - 391/(100Q))(S_{4,Q} + H_Q) \leq (c_9 + \nu - g)(S_{4,Q} + H_Q). \quad (63)$$

For the last flux interval the second inequality follows by dropping the negative correction. Substituting into $H_Q^2 = H_Q + S_{3,Q} + S_{4,Q}$ and $K_Q = \eta H_Q + K_{3,Q} + K_{4,Q}$ gives

$$\begin{aligned} H_Q^2 &\geq K_Q + AH_Q + BS_{4,Q}, \\ A &= 1 - \eta - c_9 - \nu + g - L_3 - L_4, \\ B &= \mu - \nu + g - L_3 - 2L_4. \end{aligned} \quad (64)$$

For every table row, exact rational comparison gives $A > 1/3$ and $B > 1/100$. The smallest margin $A - 1/3$ is greater than $42/10^8$. Since $K_Q, S_{4,Q} \geq 0$, this proves (60). Every estimate is on fixed few-body coefficients before the positive many-body lift, so the flux threshold is independent of total N .

At flux $Q = 2(N - 1)$ the zero-mode factor Δ_N^2 exhausts the single-particle degree, so the residual symmetric polynomial is constant. The kernel is therefore the one-dimensional Laughlin line, and the claimed ground-state gap follows by spectral calculus. In particle sectors with no zero mode, (60) means that the spectrum avoids $(0, 1/3)$; it does not assert the spacing between the two lowest positive levels.

Finally fix a plane particle number and total orbital degree. For all large Q that complete finite-dimensional sector embeds into the sphere sector and its Hamiltonian matrices converge to the plane matrices. Passing (60) to this finite-matrix limit gives the same constant. The direct sum over all degree sectors and then all particle numbers proves the plane form inequality. $\square$

8 Reproducibility and scope

The frozen certificate was replayed in a fresh Python process with site packages disabled. All 26 blocks passed the exact matrix, rank, and kernel checks. An independent implementation reconstructing normalized occupation-space operators from the frozen rows agreed with the exact two-body cost and all blocks. Separate checks verified the lift normalization $\mathcal{L}_r^N(I) = \binom{N}{r}I$, normal ordering through five-particle sectors, CM descendant metrics, and the finite- Q recoupling formula. The coherent-frame constants and their normalizations were also cross-checked.

A separate implementation reconstructed the four-body auxiliary kernel directly from the total raising matrix. For $a = 1/16$, all descendants contributing to the pair-north Schur sum were included at $Q = 32, 64, 128$, for $z = 0, 1, 2, 4, 8, 16$. At $z = 16$ the smallest eigenvalues

were respectively 0.101477757, 0.114709929, and 0.120986698, approaching the predicted limit 0.127080285. Setting the coherent weight exponent to zero recovered the identity frame with maximum residual 1.1×10^{-13} . Tensor–Fock normalization and the coherent double integral were also checked independently. These checks used floating-point arithmetic.

Separately, the frozen rows were evaluated with the actual spherical pair coefficients and normalized $SU(2)$ descendants. For each retained block define the energy-relative residual

$$e_{r,z,Q} := \lambda_{\min} \left[H_{r,z,Q}^{-1/2} E_{r,z,Q} H_{r,z,Q}^{-1/2} \big|_{\text{ran } H_{r,z,Q}} \right], \quad (65)$$

where $H_{r,z,Q}$ and $E_{r,z,Q}$ are the block restrictions and the inverse acts only on the positive image of $H_{r,z,Q}$. In double precision, the minimum over all retained nonzero blocks was

Q	$\min_{r,z} e_{r,z,Q}$
10^3	-7.42874×10^{-4}
10^5	-6.94088×10^{-6}
10^7	-3.58164×10^{-7}
10^9	-3.58164×10^{-7}
Plane	-3.58164×10^{-7}

Kernel and descendant-metric residuals were below 1.1×10^{-15} and 1.8×10^{-15} , respectively. The frozen head certificate therefore does not numerically fit the chosen allowance 2×10^{-6} at $Q = 10^5$; it does at the two larger sampled fluxes. The original conservative threshold in Appendix A follows from uniform analytic estimates and exact rational comparisons.

The verifier requires the shifted matrix to be positive semidefinite, to annihilate the Hamiltonian kernel, and to have the same rank as the Hamiltonian. The frozen rows pass all three checks.

This candidate proof is accompanied by the finite data and a replayable checker. It has not been checked by an external referee or formalized in a proof assistant. The explicit sufficient threshold $Q_0 = 3088$ follows from the interval certificates and exact budgets in Appendix B. No optimality is claimed for this threshold, which does not locate an onset of the physical gap. At Laughlin flux this sufficient range starts at $N = 1545$. The result concerns the complete projected LLL interaction on the round sphere and its planar limit.

Source. The Fock-space comparison is adapted from OpenAI [7], which treats the fermionic V_1 interaction. The bosonic coherent-error bounds and frozen rows used here are given above.

A The original conservative threshold

This appendix gives the original conservative bound 10^{50} ; Appendix B gives the current threshold 3088. The original argument assigns an explicit sufficient threshold to each large- Q step in the gap inequality. All comparisons are made on fixed few-body spaces before positive lifting, so their thresholds are uniform in the eventual particle number. The three thresholds are

Ingredient	Sufficient integer flux
Negative three-body tail with allowance $1/2000$	$Q \geq 10^7$
Finite coherent comparison bounds 35, 2100	$Q \geq 10^{50}$
Frozen head certificate with allowance 2×10^{-6}	$Q \geq 10^{50}$

Their common sufficient threshold is therefore $Q_0 = 10^{50}$. This value comes from the loose norm estimates below. The command

```
python -S verify_explicit_threshold.py
```

replays the frozen shifted certificate, the finite planar half-gap, and all scalar budgets using Python's standard library. The analytic identities and norm estimates in this appendix complete the argument.

A.1 An explicit threshold for the negative three-body tail

The following estimate replaces the existential threshold of Theorem 2 at the values needed for the gap budget.

Lemma 2. *For every integer $Q \geq 10^7$, with the full spherical pair Hamiltonian normalized as in Section 1,*

$$T_{9,Q} \leq \left(c_9 + \frac{1}{2000}\right) (S_{4,Q} + H_Q), \quad c_9 = 30 \left(\frac{5}{9}\right)^9 = \frac{19531250}{129140163}. \quad (66)$$

Proof. Put

$$a = \frac{2}{27}, \quad M = \lfloor aQ \rfloor, \quad r = \frac{9}{10}, \quad t = \frac{9}{29}, \quad \rho = \frac{29}{36}. \quad (67)$$

On the auxiliary space $V_Q \otimes U_Q$, write Π_z for the orthogonal spin- $(3Q/2 - z)$ projection, $0 \leq z \leq Q$. Recall the positive coherent operator

$$A_{M,Q} = (2Q+1)(Q+1) \int d\Omega_\alpha d\Omega_\beta s(\alpha, \beta)^M |\alpha_{\text{pair}}\rangle \langle \alpha_{\text{pair}}| \otimes |\beta\rangle \langle \beta| = \sum_{z=0}^Q \tau_{z,Q} \Pi_z. \quad (68)$$

Here each sphere measure has total mass one, the coherent vectors are normalized, and $s(\alpha, \beta) = \cos^2(\theta_{\alpha\beta}/2)$. The exact eigenvalue formula is

$$\begin{aligned} \tau_{z,Q} &= \frac{2Q+1}{3Q-2z+1} \sum_{j=z}^Q b_{j,Q} \gamma_{z,j,Q}, \\ b_{j,Q} &= \frac{Q+1}{M+Q+1} \frac{(Q)_j}{(M+Q)_j}, \\ \gamma_{z,j,Q} &= \frac{(2Q)_z}{(3Q-z+1)_z} \binom{j}{z} \frac{(Q-z)_{j-z}}{(3Q-2z)_{j-z}}. \end{aligned} \quad (69)$$

The notation $(x)_k = x(x-1)\cdots(x-k+1)$ denotes a descending factorial, with $(x)_0 = 1$. Every term in this sum is nonnegative, and the $j=z$ term is positive.

We first bound the finite set $0 \leq z \leq 32$ by retaining $j \leq 100$. For integers $0 \leq z \leq j \leq 100$, define

$$\ell_{z,j} := \left(\frac{2}{3}\right)^{z+1} (1+a)^{-j-1} \binom{j}{z} \left(\frac{1}{3}\right)^{j-z} = 2^{z+1} \binom{j}{z} t^{j+1}. \quad (70)$$

For $Q \geq 100$, the j th summand in (69), including the factor outside the sum, is at least $\ell_{z,j} P_{z,j,Q}$, where

$$P_{z,j,Q} = \prod_{i=0}^{j-1} \left(1 - \frac{i}{Q}\right) \prod_{i=0}^{z-1} \left(1 - \frac{i}{2Q}\right) \prod_{i=z}^{j-1} \left(1 - \frac{i}{Q}\right). \quad (71)$$

To see this factor by factor, use

$$\frac{2Q+1}{3Q-2z+1} \geq \frac{2}{3}, \quad \frac{Q+1}{M+Q+1} \geq \frac{1}{1+a}, \quad \frac{Q-i}{M+Q-i} \geq \frac{1-i/Q}{1+a}. \quad (72)$$

For $0 \leq i < z$ and $z \geq 1$, $(2Q-i)/(3Q-z+1-i) \geq (2/3)(1-i/(2Q))$; for $z=0$ this product is empty. For $0 \leq i < j-z$, $(Q-z-i)/(3Q-2z-i) \geq (1/3)(1-(z+i)/Q)$. All denominators are positive in the stated range. These inequalities prove (71).

Every factor in that product belongs to $[0, 1]$. The elementary inequality $\prod_k (1 - u_k) \geq 1 - \sum_k u_k$ for $0 \leq u_k \leq 1$, proved by induction, therefore gives

$$P_{z,j,Q} \geq 1 - \frac{j(j-1) - z(z-1)/4}{Q} \geq 1 - \frac{9900}{Q}. \quad (73)$$

Consequently, for $0 \leq z \leq 32$ and $Q \geq 100$,

$$\tau_{z,Q} \geq \left(1 - \frac{9900}{Q}\right) \sum_{j=z}^{100} \ell_{z,j}. \quad (74)$$

The binomial-series identity, with $0 < t < 1$, evaluates the full positive sum:

$$\sum_{j=z}^{\infty} \ell_{z,j} = \left(\frac{2t}{1-t}\right)^{z+1} = r^{z+1}. \quad (75)$$

Since $\binom{j}{z} \leq 2^j$ and $2t < 1$, its omitted part has the uniform relative bound

$$\begin{aligned} r^{-z-1} \sum_{j=101}^{\infty} \ell_{z,j} &\leq \left(\frac{2}{r}\right)^{z+1} \frac{t}{1-2t} (2t)^{101} \\ &\leq \left(\frac{20}{9}\right)^{33} \frac{9}{11} \left(\frac{18}{29}\right)^{101} \leq \frac{1}{1000}, \quad 0 \leq z \leq 32. \end{aligned} \quad (76)$$

The checker specified below verifies the last rational inequality exactly. For every $Q \geq 10^7$, $9900/Q \leq 1/1000$. Thus (74)–(76) imply

$$\tau_{z,Q} \geq \left(\frac{999}{1000}\right)^2 r^{z+1} \geq \frac{499}{500} r^{z+1}, \quad 0 \leq z \leq 32. \quad (77)$$

Since $(Q-i)/(2Q-i) \leq 1/2$ for $0 \leq i < z \leq Q$, $f_{z,Q} = (Q)_z/(2Q)_z \leq 2^{-z}$. Define

$$A_* := \frac{2}{r} \left(\frac{1}{2r}\right)^9 = \frac{20}{9} \left(\frac{5}{9}\right)^9 = ac_9. \quad (78)$$

Since $1/(2r) = 5/9 < 1$, (77) gives

$$\frac{2f_{z,Q}}{\tau_{z,Q}} \leq \frac{500}{499} A_*, \quad 9 \leq z \leq 32. \quad (79)$$

The remaining range is controlled without any expansion in z/Q . Keeping only the $j = z$ term in (69) gives

$$\begin{aligned} \frac{2f_{z,Q}}{\tau_{z,Q}} &\leq 2 \frac{3Q-2z+1}{2Q+1} \frac{M+Q+1}{Q+1} \prod_{i=0}^{z-1} \frac{(M+Q-i)(3Q-z+1-i)}{(2Q-i)^2} \\ &\leq 3(1+a)\rho^z. \end{aligned} \quad (80)$$

For the last inequality the prefactor is at most $3(1+a)$. With $x = i/Q \in [0, 1]$, each product factor is bounded by

$$\frac{(1+a-x)(3-2x)}{(2-x)^2} \leq \frac{3(1+a)}{4} = \rho. \quad (81)$$

After multiplication by the positive denominator, the difference is $x[(8-4a) - (5-3a)x]/4 \geq 0$. The second finite rational comparison is

$$3 \frac{29}{27} \left(\frac{29}{36}\right)^{33} \leq A_*. \quad (82)$$

Since $0 < \rho < 1$, this proves $2f_{z,Q}/\tau_{z,Q} \leq A_*$ for every $33 \leq z \leq Q$. In conjunction with (79), it gives the whole-space inequality

$$C_{9,Q} = 2 \sum_{\substack{9 \leq z \leq Q \\ z \text{ odd}}} f_{z,Q} \Pi_z \leq \frac{500}{499} A_* A_{M,Q}, \quad Q \geq 10^7. \quad (83)$$

Apply the positive map $X \mapsto \mathcal{L}_3(W_3 X W_3^\dagger)$. The exact coherent marginal and pair frame give

$$\mathcal{L}_3(W_3 A_{M,Q} W_3^\dagger) \leq \frac{Q+1}{M+1} (S_{4,Q} + H_Q). \quad (84)$$

Because $M+1 > aQ$, for $Q \geq 1000$,

$$\frac{Q+1}{M+1} \leq \frac{1}{a} \left(1 + \frac{1}{Q}\right) \leq \frac{1}{a} \frac{1001}{1000}. \quad (85)$$

Using $A_* = ac_9$, the resulting coefficient is at most $(1001/998)c_9$. Its exact distance below the coefficient in (66) is

$$c_9 + \frac{1}{2000} - \frac{1001}{998} c_9 = \frac{1949063779}{42960627558000} > 0. \quad (86)$$

This proves the claim for every integer $Q \geq 10^7$. $\square$

The comparisons in (76), (82), and (86), together with the parameter identities and threshold inequalities, are replayed using integers and `fractions.Fraction` by `verify_explicit_tail_threshold.py` with Python's `-S` flag. The same argument also permits $Q \geq 9900000$; the rounded threshold 10^7 is used for simplicity.

A.2 An explicit threshold for the coherent comparison bounds

We make the coherent comparison bounds of Lemma 1 quantitative, independently of the finite certificate.

Lemma 3. *For every integer $Q \geq 10^{50}$, on every particle-number sector,*

$$\mathcal{R}_{3,8,Q} \leq 35(S_{4,Q} + H_Q), \quad \mathcal{R}_{4,16,Q} \leq 2100(2S_{4,Q} + H_Q). \quad (87)$$

Proof. Put $r \in \{3, 4\}$ and $h = r - 2$. The auxiliary spin space is $V_Q \otimes \text{Sym}^h U_Q$: it contains the maximal-spin pair and h spectators, symmetrized only among themselves. Introduce an auxiliary oscillator space with pair mode u and spectator modes $v_1, \dots, v_h$, restricted to spectator symmetry. Its degree- l subspace has normalized occupation labels $(p; j_1, \dots, j_h)$ with $p + \sum_s j_s = l$. At $l \leq Q$ these labels identify its orthonormal basis with the spherical auxiliary weight- l basis. All finite-weight matrix norms below use this unitary identification.

Define the center-of-mass annihilator $a_\zeta = (\sqrt{2}a_u + \sum_s a_{v_s})/\sqrt{r}$. For $z \leq 16$, let $F_{z,\infty}$ be an isometric frame from a fixed multiplicity coordinate space onto the degree- z kernel of a_ζ . Its dimension is one for $r = 3$, and $\lfloor z/2 \rfloor + 1$ for $r = 4$. We need only $z \leq 8$ when $r = 3$. On degree l , define the positive diagonal matrix

$$D_{Q,l}(p; j_1, \dots, j_h) = \sqrt{\frac{(2Q)_p}{(2Q)^p}} \prod_{s=1}^h \sqrt{\frac{(Q)_{j_s}}{Q^{j_s}}}. \quad (88)$$

It preserves spectator symmetry. With $J_\pm$ the total auxiliary spin raising and lowering operators, the exact matrix identity is

$$\frac{J_+}{\sqrt{Q}} = D_{Q,l-1}^{-1}(\sqrt{r} a_\zeta) D_{Q,l}. \quad (89)$$

Indeed, the spin raising coefficient for a factor of capacity cQ is $\sqrt{j(cQ - j + 1)}$, where $c = 2$ for the pair and $c = 1$ for a spectator; consecutive entries in (88) have ratio $\sqrt{1 - (j - 1)/(cQ)}$. Thus $D_{Q,z}^{-1}$ maps the oscillator highest space bijectively onto the spherical highest space. Normalize that map:

$$V_{z,Q} = D_{Q,z}^{-1} F_{z,\infty}, \quad G_{z,Q} = V_{z,Q}^\dagger V_{z,Q}, \quad F_{z,Q} = V_{z,Q} G_{z,Q}^{-1/2}. \quad (90)$$

Here $G_{z,Q}$ acts in the fixed multiplicity coordinate space. For $z \leq 16$, the inequality $\prod_i (1 - x_i) \geq 1 - \sum_i x_i$, for $x_i \in [0, 1]$, gives

$$(1 - 128/Q)I \leq D_{Q,z}^2 \leq I. \quad (91)$$

The sum of the factors $i/(cQ)$ is at most $z(z - 1)/(2Q) \leq 120/Q$. For $Q \geq 256$ this implies

$$\|D_{Q,z}^{-1} - I\| \leq 256/Q, \quad \|G_{z,Q}^{-1/2} - I\| \leq 256/Q, \quad \|F_{z,Q} - F_{z,\infty}\| \leq 512/Q. \quad (92)$$

For the middle estimate, use $I \leq G_{z,Q} \leq (1 - 128/Q)^{-1}I$ and functional calculus. The last estimate follows by adding and subtracting $F_{z,\infty}G_{z,Q}^{-1/2}$.

We next compare normalized descendants through $K = 1024$; set $B = 4096$. All retained weights obey $z + n \leq 1040 < B$. Define isometries from the multiplicity coordinate space to degree $z + n$ by

$$X_{z,n,Q} = \frac{J_-^n F_{z,Q}}{\sqrt{n!(rQ - 2z)_n}}, \quad X_{z,n,\infty} = \frac{(a_\zeta^\dagger)^n F_{z,\infty}}{\sqrt{n!}}. \quad (93)$$

The first normalization is the exact spin- $rQ/2 - z$ ladder normalization. The second is isometric because the highest vectors have center-of-mass occupation zero. On weight $l = z + n$, set $T_Q = J_-/\sqrt{Q}$ and $T_\infty = \sqrt{r} a_\zeta^\dagger$. The respective step maps $X_{z,n+1} = C_{z,n} X_{z,n}$ are

$$C_{z,n,Q} = \frac{T_Q}{\sqrt{r(n+1)}\sqrt{1 - x_{z,n,Q}}}, \quad C_{z,n,\infty} = \frac{T_\infty}{\sqrt{r(n+1)}}, \quad x_{z,n,Q} = \frac{2z + n}{rQ}. \quad (94)$$

In the occupation basis T_Q has nonnegative entries bounded entrywise by those of T_∞ : each one-mode coefficient acquires a factor $\sqrt{1 - i/(cQ)} \leq 1$. For such matrices, $|T_Q v| \leq T_\infty |v|$ componentwise, which implies the norm bound. Since $\|T_\infty\| = \sqrt{r(l+1)}$ on degree l ,

$$\|C_{z,n,Q}\| \leq \sqrt{\frac{z+n+1}{n+1}} (1 - x_{z,n,Q})^{-1/2}. \quad (95)$$

For $Q \geq 2^{24}$, the elementary inequality $1 - \sqrt{1 - y} \leq y$ bounds the sum of the one-mode differences by $\|T_Q - T_\infty\| \leq 3B^{3/2}/Q$. Using $(1 - x)^{-1/2} \leq 2$, $(1 - x)^{-1/2} - 1 \leq 2x$ for $0 \leq x \leq 1/2$, and $\|C_{z,n,\infty}\| \leq \sqrt{17} < 5$, gives

$$\|C_{z,n,Q} - C_{z,n,\infty}\| \leq \frac{6B^{3/2} + 10B}{Q} < \frac{2^{22}}{Q}. \quad (96)$$

To control accumulated errors, for $0 \leq s \leq n \leq K$ the factors in (95) telescope:

$$\prod_{t=s}^{n-1} \sqrt{\frac{z+t+1}{t+1}} = \sqrt{\frac{\binom{z+n}{z}}{\binom{z+s}{z}}} \leq (z+n)^{z/2} \leq 2^{88}. \quad (97)$$

Empty products equal one, including the case $z = n = 0$. Moreover, $\sum_{t=s}^{n-1} x_{z,t,Q} \leq K(32 + K)/(3Q) < 2^{19}/Q \leq 1/32$. The product inequality used above therefore bounds the product of the curvature factors $(1 - x_{z,t,Q})^{-1/2}$ by 2. Every consecutive product of the finite- Q step maps has norm at most 2^{89} . Subtracting the two descendant recursions and unrolling them, (92) and (96) now give

$$\|X_{z,n,Q} - X_{z,n,\infty}\| \leq \frac{2^{89}(512 + K2^{22})}{Q} < \frac{2^{122}}{Q}, \quad 0 \leq n \leq K. \quad (98)$$

Take $a = 1/12$ for $r = 3$, and $a = 1/16$ for $r = 4$, with $M = \lfloor aQ \rfloor$. The one-spectator north-pole seed R_Q has diagonal entries

$$b_{j,Q} = \frac{Q+1}{M+Q+1} \prod_{i=0}^{j-1} \frac{Q-i}{M+Q-i}, \quad b_{j,\infty} = (1+a)^{-j-1}. \quad (99)$$

Write $M = aQ - e$, $0 \leq e < 1$. For $j \leq B$ and $Q \geq 2^{24}$,

$$\left| \frac{Q+1}{M+Q+1} - \frac{1}{1+a} \right| \leq \frac{2}{Q}, \quad \left| \frac{Q-i}{M+Q-i} - \frac{1}{1+a} \right| \leq \frac{2(i+1)}{Q}. \quad (100)$$

For the second estimate the difference has numerator $(e-ai)/Q$ and denominator $(1+a)[1+a-(e+i)/Q] \geq 1/2$. All product factors lie in $[0, 1]$, so telescoping the products gives $|b_{j,Q} - b_{j,\infty}| \leq [2+j(j+1)]/Q < 2^{26}/Q$. On degree l , define the pair-north seeds

$$\Sigma_{Q,l} = \left[|0_u\rangle\langle 0_u| \otimes (R_Q^{\otimes h})_{\text{sym}} \right]_l, \quad \Sigma_{\infty,l} = \left[|0_u\rangle\langle 0_u| \otimes (R_{\infty}^{\otimes h})_{\text{sym}} \right]_l. \quad (101)$$

Brackets with subscript l mean restriction to degree l . The symmetric spectator basis is orthonormal also for coincident occupations, so no extra combinatorial factor enters these diagonal operators. On the retained weights,

$$\|\Sigma_{Q,l} - \Sigma_{\infty,l}\| \leq 2^{27}/Q, \quad \|\Sigma_{Q,l}\|, \|\Sigma_{\infty,l}\| \leq 1. \quad (102)$$

For the remaining weights, $Q \geq 2/a$ gives $M \geq aQ/2$ and hence $0 \leq b_{j,Q} \leq (1+a/2)^{-j}$. For both choices of a , put $q = 32/33$; then $(1+a/2)^{-1} \leq q$, and the limiting seed obeys the same bound. The pair-north slice at degree l has spectator degree l , giving

$$\|\Sigma_{Q,l}\|, \|\Sigma_{\infty,l}\| \leq q^l. \quad (103)$$

If the finite pair-north slice is absent, its seed is zero.

The exact Schur formula for the spin-deficit- z multiplicity matrix is

$$A_{r,z,Q} = c_{r,z,Q} \sum_{n=0}^{rQ-2z} X_{z,n,Q}^\dagger \Sigma_{Q,z+n} X_{z,n,Q}, \quad c_{r,z,Q} = \frac{2Q+1}{rQ-2z+1}. \quad (104)$$

The oscillator matrix $A_{r,z,\infty}$ is the same sum over $n \geq 0$ with the limiting frames and seeds, and $c_{r,z,\infty} = 2/r$. This is the center-of-mass partial trace evaluated by the displacement twirl in the proof of Lemma 1. For $Q \geq 2^{24}$ and $z \leq 16$, $c_{r,z,Q}, c_{r,z,\infty} \leq 1$ and $|c_{r,z,Q} - c_{r,z,\infty}| \leq 32/Q$. Both descendant frames are isometries, so a compressed-seed difference is bounded by twice the frame error plus the seed error. Thus the truncated Schur sums differ by at most

$$\frac{(K+1)(32+2^{27}+2^{123})}{Q} < \frac{2^{135}}{Q}. \quad (105)$$

The sum of their omitted tails is, by (103), at most $2q^{K+1}/(1-q) = 66(32/33)^{1025}$. The binomial theorem gives $(33/32)^{32} > 2$, and therefore

$$\|A_{r,z,Q} - A_{r,z,\infty}\| \leq \frac{2^{135}}{Q} + 66(32/33)^{1025} < \frac{2^{135}}{Q} + \frac{66}{2^{32}} < 10^{-6}, \quad Q \geq 10^{50}. \quad (106)$$

This is uniform in the retained z and all multiplicities.

For $r = 3$, the exact limiting eigenvalue is $[2/(2+3a)]^{z+1}$, whose minimum for $a = 1/12$, $z \leq 8$ is $t_3 = (8/9)^9$. For $r = 4$, the eigenvalues are $(1+a)^{-n_1-1}(1+2a)^{-n_2-1}$, with $n_1+n_2 = z$ and n_1 even; their minimum for $a = 1/16$, $z \leq 16$ is $t_4 = (16/17)(8/9)^{17}$. Put $\delta = 10^{-6}$. Equation (106), invariance, and positivity outside the retained spin blocks imply the whole-space inequalities

$$\Pi_{\leq D,Q}^{\text{aux}} \leq (t_r - \delta)^{-1} A_{r,Q}, \quad (r, D) = (3, 8), (4, 16). \quad (107)$$

Here $A_{3,Q} = A_{M,Q}$ and $A_{4,Q} = A_{M,Q}^{(4)}$ are precisely the positive kernels of the preceding proofs. Compose with $W_r, W_r^\dagger$ and use the positive lift and exact spectator marginals. Since $(Q+1)/(M+1) \leq a^{-1}(1+1/Q)$ and $1+1/Q \leq 10001/10000$ for $Q \geq 10^{50}$, the resulting coefficients are bounded by the exact rational numbers

$$\frac{12(10001/10000)}{(8/9)^9 - 10^{-6}} < 35, \quad \frac{256(10001/10000)^2}{(16/17)(8/9)^{17} - 10^{-6}} < 2100. \quad (108)$$

All comparisons precede the many-body lift, so the threshold is independent of particle number. The companion verifier `check_explicit_coherent_threshold.py` checks the final rational inequalities and the integer bounds used in these constants exactly. $\square$

A.3 An explicit threshold for transferring the finite certificate

The estimates in this subsection give the uniform threshold $Q_{\text{head}} = 10^{50}$ for the finite comparison errors. They use an additional exact calculation on the finite planar relative spaces retained in the certificate:

$$H_{r,\infty}^2 - \frac{1}{2}H_{r,\infty} \geq 0 \quad \text{on relative degree } z \leq D_r, \quad D_3 = 8, \quad D_4 = 16. \quad (109)$$

The companion `exact_planar_head_gap.py` verifies all 26 rational quadratic forms and checks that each relative space is invariant under $H_{r,\infty}$. Since $H_{r,\infty} \geq 0$, its nonzero eigenvalues on these finite spaces are therefore at least $1/2$. Throughout this subsection $Q \geq 1024$, and all norms are operator norms in normalized occupation coordinates.

For $n \leq 4$ and total orbital weight $M \leq 16$, let $\mathcal{W}_{n,M}$ be the finite coordinate Hilbert space with orthonormal basis indexed by partitions $A = (j_1, \dots, j_n)$ with $\sum_i j_i = M$. These are complete fixed-weight physical spaces: the same labels identify the spherical and planar normalized occupation bases for $Q \geq 16$. Define the invertible positive diagonal map $D_{n,M,Q} : \mathcal{W}_{n,M} \rightarrow \mathcal{W}_{n,M}$ by

$$D_{n,M,Q}|A\rangle = d_A|A\rangle, \quad d_A = \prod_{j \in A} \sqrt{\prod_{\ell=0}^{j-1} (1 - \ell/Q)}. \quad (110)$$

It converts normalized spherical coefficients to coefficients of the same polynomial in the planar monomial convention: after writing $x = \sqrt{Q}v/u$ and removing u^Q , the spherical orbital is $d_j x^j / \sqrt{j!}$. The map is not unitary. If J_+ decreases spherical orbital weight and L is the planar CM lowering operator, direct action on the basis gives the exact identity

$$Q^{-1/2}J_+ = D_{n,M-1,Q}^{-1}LD_{n,M,Q}. \quad (111)$$

For the pair annihilator from $\mathcal{W}_{n,M}$ to $\mathcal{W}_{n-2,M-p}$, the exact Clebsch–Gordan coefficients similarly give

$$B_{p,Q} = e_{p,Q}^{-1}D_{n-2,M-p,Q}^{-1}B_{p,\infty}D_{n,M,Q}, \quad e_{p,Q} = \sqrt{\prod_{\ell=0}^{p-1} (1 - \ell/(2Q))}. \quad (112)$$

The zero-particle diagonal map is the identity.

Fix r, z in the retained ranges, and let $U : \mathbb{C}^m \rightarrow \mathcal{W}_{r,z}$ be any isometric frame for $\ker L$, where $m = \dim \ker L$. The map $V_{Q,0} = D_{r,z,Q}^{-1}U$ is an invertible frame onto the spherical highest-weight space, by (111). Pullback by $V_{Q,0}$ is a congruence of quadratic forms. Equation (112) shows that their Hamiltonian kernel is the fixed subspace

$$\mathcal{N} := \ker(U^\dagger H_{r,\infty} U) = \ker(V_{Q,0}^\dagger H_{r,Q} V_{Q,0}) \subset \mathbb{C}^m. \quad (113)$$

Indeed both kernels are the simultaneous kernels of all corresponding pair annihilators. The normal-ordered pair-sandwich structure of K_r , and invariance of $\ker H_r$ under rotations and spin

projections, imply that the congruent E_r matrices also annihilate $\mathcal{N}$ exactly. Hermiticity then excludes matrix elements between $\mathcal{N}$ and $\mathcal{N}^\perp$.

Since $\sum_{j \in A} j(j-1)/2 \leq 120$ at weight at most 16, the elementary product inequality $\prod_i (1 - x_i) \geq 1 - \sum_i x_i$ gives

$$\|D - I\| \leq 120/Q, \quad \|D^{-1}\| \leq 2, \quad \|D^{-1} - I\| \leq 240/Q, \quad |e_p^{-1} - 1| \leq 120/Q. \quad (114)$$

Each pair interaction is a projector, so $0 \leq H_n \leq \binom{n}{2}I \leq 6I$ and $\|B_p\| < 3$ on either geometry. Subtracting $B_{p,\infty}$ from (112) and using (114) therefore gives

$$\|B_{p,Q} - B_{p,\infty}\| \leq \frac{120 \cdot 2 \cdot 3 + 240 \cdot 3 + 120 \cdot 3}{Q} < \frac{2000}{Q}, \quad \|H_Q - H_\infty\| < \frac{10^6}{Q}. \quad (115)$$

The second estimate sums at most 17 pair channels. For the target $S_3 = H_3^2 - H_3$, its norm is at most 42 and its difference is at most $13 \cdot 10^6/Q$. For S_4 on weight M , set $T_p = B_{M-p}^{(2)} B_p^{(4)}$. There are at most 17 such terms, $\|T_p\| \leq 9$, and $\|T_{p,Q} - T_{p,\infty}\| \leq 12000/Q$. Thus, writing $\mathbb{T}_3 = S_3$ and $\mathbb{T}_4 = (1 - c_9 - \mu)S_4$, where $0 < 1 - c_9 - \mu < 1$, the following common bounds hold:

$$\|\mathbb{T}_{r,Q}\|, \|\mathbb{T}_{r,\infty}\| < 2000, \quad \|\mathbb{T}_{r,Q} - \mathbb{T}_{r,\infty}\| < 10^8/Q. \quad (116)$$

The frozen JSON has 138 rows, at most 46 coefficients in each row, and all its scaled coefficients y have absolute value less than one. The scaling (52), with $p, j, i \leq 8$, yields $|\lambda| \leq 2$ and $|\alpha_{pj}| \leq \sqrt{2^9 8!} < 5000$. On the output of a pair annihilator at most two particles remain, and $\|a_i^\dagger a_j\| \leq 2$ there. This last bound follows by expressing the operator as a sum of the single-particle operator $|i\rangle\langle j|$ in the remaining particle slots. Consequently

$$\|F_Q\|, \|F_\infty\| < 2 \cdot 10^6, \quad \|F_Q - F_\infty\| < 10^9/Q. \quad (117)$$

For $R_n = \sum_\rho F_\rho^\dagger F_\rho|_n$, these imply $\|R_n\| < 10^{15}$ and $\|R_{n,Q} - R_{n,\infty}\| < 10^{18}/Q$. The normalized k -body lift on n particles is the sum over unordered k -particle subsets, hence has norm as a linear map on bounded operators at most $\binom{n}{k}$. It is the same map in matched occupation coordinates. For the Hermitian inputs needed here, this bound also follows directly from positivity and $\mathcal{L}_k^n(I) = \binom{n}{k}I$, applied to $- \|R\|I \leq R \leq \|R\|I$. Apply this to the recursive normal-order extraction of K_r . The respective multipliers for $r = 2, 3, 4$ are 1, $1 + 3 = 4$, and $1 + 6 + 4 \cdot 4 = 23$. For every retained unaveraged body-weight matrix we therefore obtain

$$\|K_{r,M,Q}\|, \|K_{r,M,\infty}\| < 3 \cdot 10^{16}, \quad \|K_{r,M,Q} - K_{r,M,\infty}\| < 3 \cdot 10^{19}/Q. \quad (118)$$

Here body-weight matrices are extended by zero before applying a lift; the norm of their direct sum is their maximum norm.

For $k = M - z$, define the descendant frames recursively by

$$V_{Q,k} = T_{Q,k} V_{Q,k-1}, \quad T_{Q,k} := \frac{J_- / \sqrt{Q}}{\sqrt{k[r - (2z + k - 1)/Q]}}. \quad (119)$$

Let U_k be the analogous planar frame, starting at U and replacing $T_{Q,k}$ by $T_{\infty,k} = L^\dagger / \sqrt{rk}$. The $SU(2)$ ladder identity gives $V_{Q,k}^\dagger V_{Q,k} = V_{Q,0}^\dagger V_{Q,0}$ exactly, so $\|V_{Q,k}\| \leq 2$, while $\|U_k\| = 1$. On weight $M - 1$, $\|L^\dagger\| \leq \sqrt{rM} \leq 8$. Every nonnegative spherical matrix entry is the corresponding planar entry times $\sqrt{1 - j/Q}$ for $j \leq 15$, giving $\|J_- / \sqrt{Q} - L^\dagger\| \leq 120/Q$ by entrywise domination. The inverse denominator in (119) differs from $1/\sqrt{rk}$ by at most $11/Q$, since $(2z + k - 1)/(rQ) \leq 31/(3Q)$. The resulting uniform bounds are

$$\|T_{Q,k}\| \leq 16, \quad \|T_{Q,k} - T_{\infty,k}\| \leq 512/Q. \quad (120)$$

With $\epsilon_k = \|V_{Q,k} - U_k\|$, we have $\epsilon_0 \leq 240/Q$ and $\epsilon_k \leq 16\epsilon_{k-1} + 512/Q$. Since $k \leq 16$,

$$\epsilon_k \leq \frac{240 \cdot 16^{16} + 512(16^{16} - 1)/15}{Q} < \frac{10^{22}}{Q}. \quad (121)$$

The exact Schur factor is $f_Q = (2 + 1/Q)/(r + (1 - 2z)/Q)$; its limit is $f_\infty = 2/r$. For the specified ranges, $0 < f_Q \leq 1$ and $|f_Q - f_\infty| \leq 16/Q$. If a tilde denotes pullback of a relative quadratic form by $V_{Q,0}$, the averaged coefficient is exactly

$$\tilde{K}_{r,z,Q} = f_Q \sum_{M=z}^{D_r} V_{Q,M-z}^\dagger K_{r,M,Q} V_{Q,M-z}. \quad (122)$$

This identity holds for the nonorthonormal highest frame because the normalized ladder maps preserve its Gram matrix. Combining (118) and (121), each summand differs from its planar counterpart by at most $[4(3 \cdot 10^{19}) + 3(3 \cdot 10^{16})10^{22}]/Q$. There are at most 17 summands, and the Schur-factor error contributes at most $16 \cdot 17 \cdot 3 \cdot 10^{16}/Q$. Hence

$$\|\tilde{K}_{r,z,Q} - \tilde{K}_{r,z,\infty}\| < 10^{42}/Q. \quad (123)$$

For the highest-frame targets, (114) gives the bounds $[4 \cdot 10^8 + 3 \cdot 2000 \cdot 240]/Q < 10^9/Q$ and $[4 \cdot 10^6 + 3 \cdot 6 \cdot 240]/Q < 5 \cdot 10^6/Q$ for the target and Hamiltonian differences, respectively. Thus with $\delta = 10^{-6}$ and $\tilde{F}_Q := \tilde{E}_{r,z,Q} + 2\delta\tilde{H}_{r,z,Q}$,

$$\|\tilde{F}_Q - \tilde{F}_\infty\| \leq 10^{43}/Q. \quad (124)$$

The frozen certificate (54) and (109) imply, on the fixed $\mathcal{N}^\perp$,

$$\tilde{F}_\infty = (\tilde{E}_\infty + \delta\tilde{H}_\infty) + \delta\tilde{H}_\infty \geq \frac{\delta}{2}I = 5 \cdot 10^{-7}I. \quad (125)$$

At every integer $Q \geq 10^{50}$, the error in (124) is at most 10^{-7} . The remaining margin on $\mathcal{N}^\perp$ is therefore at least $4 \cdot 10^{-7}$, while $\tilde{F}_Q$ annihilates $\mathcal{N}$ exactly. Congruence and rotational invariance give

$$E_{r,Q} \geq -2 \cdot 10^{-6} H_{r,Q} \Pi_{\leq D_r,Q}^{(r)}, \quad r = 3, 4, \quad Q \geq 10^{50}. \quad (126)$$

The companion `explicit_head_threshold_checker.py` checks all scalar budgets with integers and fractions, validates the frozen row bounds, and independently recomputes the finite planar half-gap. The planar shifted certificate itself is replayed by `exact_shifted_certificate.py`.

B The optimized threshold $Q_0 = 3088$

The frozen operator rows are unchanged. The optimized argument replaces the coarse perturbation norms of the previous appendix by direct interval certification and uses one coherent comparison coefficient for each retained spin deficit. The real kernel exponents have slopes $a_3 = 83/500$ and $a_4 = 43/500$. These parameters enter only the auxiliary coherent kernels.

For each lower endpoint q_- the verifier computes $\delta_{r,z} = L_r/\overline{C}_{r,z}(q_-)$ exactly. The coefficient $\overline{C}$ is defined below and controls every $Q \geq q_-$. The head verification likewise covers the entire interval $0 \leq 1/Q \leq 1/q_-$, including the plane endpoint. Restricting that already uniform statement to the row's flux interval permits the sharper tail budget. The overlapping intervals cover the full flux range. Six runs each contain all nine three-body and all seventeen four-body degree blocks, including zero-dimensional ones, for 156 block checks.

Run `python-Sverify_optimized_threshold.py` from the certificate directory. It reconstructs the exact budgets, checks all shifts against the frozen files, replays every interval certificate, and checks the six final rational budgets and interval coverage. It also replays the original rational planar certificate and the optimized tail checker and uses only the Python standard library.

Flux interval	L_3	L_4	$A - 1/3$ lower bound
[3088, 3090]	0.00442365	0.00280612	$4.2 \cdot 10^{-7}$
[3090, 3100]	0.00442076	0.00280405	$1.3 \cdot 10^{-6}$
[3100, 3125]	0.004409	0.002802	$5.0 \cdot 10^{-6}$
[3125, 3300]	0.004373	0.002776	$6.6 \cdot 10^{-7}$
[3300, 4000]	0.004139	0.002610	$1.93 \cdot 10^{-4}$
[4000, ∞)	0.003410	0.002100	$4.54 \cdot 10^{-4}$

Table 1: Exact terminating-decimal budgets. For finite upper endpoint q_+ use $g = 391/(100q_+)$ in (64); the last row uses $g = 0$. The displayed margin bounds are rounded down and are checked as rational inequalities. Every row has $B > 1/100$.

B.1 An optimized negative-tail bound at every flux

Allowing a real coherent weight exponent gives a negative-tail estimate valid at every flux and a finite- Q improvement.

Lemma 4. Put $c_9 = 30(5/9)^9$ and $\nu = 10^{-10}$. For every integer $Q \geq 0$,

$$T_{9,Q} \leq (c_9 + \nu)(S_{4,Q} + H_Q). \quad (127)$$

The stronger bound

$$T_{9,Q} \leq \left(c_9 + \nu - \frac{d}{Q}\right)(S_{4,Q} + H_Q) \quad (Q \geq Q_*) \quad (128)$$

holds for each of the following independent choices:

$$(Q_*, d) = (2500, 77/20), \quad (3000, 391/100), \quad (5000, 4). \quad (129)$$

All these inequalities are independent of total particle number.

Proof. Take $a = 2/27$ and $M = aQ$ exactly. The positive coherent auxiliary operator is

$$A_{M,Q} = (2Q+1)(Q+1) \int d\Omega_\alpha d\Omega_\beta s(\alpha, \beta)^M |\alpha_{\text{pair}}\rangle \langle \alpha_{\text{pair}}| \otimes |\beta\rangle \langle \beta|. \quad (130)$$

The measures have total mass one, the vectors are normalized, and $s(\alpha, \beta) = \cos^2(\theta_{\alpha\beta}/2)$. Here M is only the exponent of a nonnegative weight, not a spin label; the integral defines a positive invariant operator for every real $M \geq 0$. The exact beta integral still gives

$$b_{j,Q} = (Q+1) \binom{Q}{j} \int_0^1 s^{M+Q-j} (1-s)^j ds = \frac{Q+1}{M+Q+1} \frac{(Q)_j}{(M+Q)_j}. \quad (131)$$

This formula follows by j integrations by parts, valid for $0 \leq j \leq Q$ and $M \geq 0$. The notation $(x)_k$ denotes a descending factorial, also for real x , with $(x)_0 = 1$.

Writing $A_{M,Q} = \sum_z \tau_{z,Q} \Pi_z$ in the auxiliary pair-single spin decomposition, the exact eigenvalue formula remains

$$\begin{aligned} \tau_{z,Q} &= \frac{2Q+1}{3Q-2z+1} \sum_{j=z}^Q b_{j,Q} \gamma_{z,j,Q}, \\ \gamma_{z,j,Q} &= \frac{(2Q)_z}{(3Q-z+1)_z} \binom{j}{z} \frac{(Q-z)_{j-z}}{(3Q-2z)_{j-z}}. \end{aligned} \quad (132)$$

Every term is nonnegative and $\tau_{z,Q} > 0$. The exact coherent marginal and positive lifting give

$$\mathcal{L}_3(W_3 A_{M,Q} W_3^\dagger) \leq \frac{Q+1}{M+1} (S_{4,Q} + H_Q). \quad (133)$$

Thus it suffices to bound, on every odd $9 \leq z \leq Q$, the scalar cost

$$k_{z,Q} := \frac{Q+1}{M+1} \frac{2f_{z,Q}}{\tau_{z,Q}}, \quad f_{z,Q} = \frac{(Q)_z}{(2Q)_z}. \quad (134)$$

For $Q > 0$, put $x = 1/Q$ and $D_{z,Q} := 2/k_{z,Q} = [(M+1)/(Q+1)]\tau_{z,Q}/f_{z,Q}$. Dividing each $j = z + n$ term in (132) by $f_{z,Q}$ and cancelling powers of Q gives the exact positive sum

$$D_{z,Q} = \sum_{n=0}^{Q-z} g_{z,n}(x), \quad (135)$$

where

$$\begin{aligned} g_{z,n}(x) &= \binom{z+n}{z} \frac{(2+x)(a+x)}{[3 - (2z-1)x](1+a+x)} \\ &\quad \times \prod_{i=0}^{z-1} \frac{(2-ix)^2}{(1+a-ix)[3 - (z-1+i)x]} \\ &\quad \times \prod_{i=0}^{n-1} \frac{[1 - (z+i)x]^2}{[1+a - (z+i)x][3 - (2z+i)x]}. \end{aligned} \quad (136)$$

For $Q \geq 256$ and $z \leq 25$, retain the first 33 terms:

$$G_z(x) := \sum_{n=0}^{32} g_{z,n}(x) \leq D_{z,Q}. \quad (137)$$

This truncation is valid because $z + 32 \leq 57 < Q$. All linear factors in these rational functions are strictly positive on $[0, 1/256]$.

For a positive term $g(x) = b \prod_i (c_i + d_i x) / \prod_j (e_j + f_j x)$ on $[0, h]$, let the subscript $-$ or $+$ on a linear factor denote its minimum or maximum endpoint value. Then

$$g_- := \frac{b \prod_i (c_i + d_i x)_-}{\prod_j (e_j + f_j x)_+} \leq g(x) \leq g_+ := \frac{b \prod_i (c_i + d_i x)_+}{\prod_j (e_j + f_j x)_-}. \quad (138)$$

Since $d/(c + dx)$ has derivative $-d^2/(c + dx)^2 \leq 0$,

$$\frac{g'(x)}{g(x)} \geq L_h(g) := \sum_i \frac{d_i}{c_i + d_i h} - \sum_j \frac{f_j}{e_j}. \quad (139)$$

Hence a constant derivative lower bound is

$$\beta_h(g) := \begin{cases} L_h(g)g_-, & L_h(g) \geq 0, \\ L_h(g)g_+, & L_h(g) < 0. \end{cases} \quad g'(x) \geq \beta_h(g). \quad (140)$$

Repeated factors in (136) are simply listed repeatedly. All operations and sign decisions in these rules are rational, and the bounds hold throughout the interval.

Applying these rules to (136), the supplied exact checker verifies

$$\sum_{n=0}^{32} \beta_{1/256}(g_{9,n}) > 73, \quad (141)$$

$$G_9(0) \geq \frac{2}{c_9 + 10^{-10}}, \quad (142)$$

$$\sum_{n=0}^{32} (g_{z,n})_- > 20, \quad z = 11, 13, \dots, 25, \quad h = 1/256. \quad (143)$$

The endpoint is also checked independently through

$$G_9(0) = \frac{2}{c_9} \left(\frac{20}{29}\right)^{10} \sum_{n=0}^{32} \binom{n+9}{9} \left(\frac{9}{29}\right)^n. \quad (144)$$

Thus G_9 is increasing on the interval, so $k_{9,Q} \leq c_9 + \nu$ for every $Q \geq 256$. The other retained odd blocks satisfy $k_{z,Q} < 1/10 < c_9$ by (143).

For all remaining spins, the coarse bound obtained by retaining $j = z$ in (132) remains valid with real $M = aQ$:

$$\frac{2f_{z,Q}}{\tau_{z,Q}} \leq 3(1+a)\rho^z, \quad \rho = \frac{3(1+a)}{4} = \frac{29}{36}. \quad (145)$$

Its factor-by-factor proof uses only $M \leq aQ$, not integrality. Since $(Q+1)/(aQ+1) \leq 1/a$, every $z \geq 27$ obeys

$$k_{z,Q} \leq \frac{3(1+a)}{a} \left(\frac{29}{36}\right)^{27} < \frac{13}{100} < c_9, \quad (146)$$

with both comparisons checked exactly.

To exhaust the finite flux remainder, the checker verifies all 2151 pairs consisting of an integer $9 \leq Q \leq 255$ and an odd $9 \leq z \leq \min(Q, 25)$. For each such pair it checks

$$\sum_{n=0}^{\min(32, Q-z)} g_{z,n}(1/Q) > \frac{100}{7}. \quad (147)$$

The omitted terms in (135) are nonnegative, so these blocks have $k_{z,Q} < 7/50 < c_9$. For $Q < 9$, the negative tail is empty. These arguments prove (127).

Finally put $C = c_9 + \nu$. The same exact interval rules verify

$$\sum_{n=0}^{32} \beta_{1/Q_*}(g_{9,n}) > b \quad \text{for } (Q_*, b, d) = (2500, 681/2, 77/20), (3000, 345, 391/100), (5000, 353, 4). \quad (148)$$

For each triple the checker also verifies

$$bC^2 \geq d \left(2 + \frac{bC}{Q_*}\right), \quad C - \frac{d}{Q_*} > \frac{13}{100}. \quad (149)$$

Equations (142) and (148) imply, for every $Q \geq Q_*$,

$$D_{9,Q} \geq \frac{2}{C} + \frac{b}{Q}, \quad k_{9,Q} \leq \frac{C}{1 + bC/(2Q)} \leq C - \frac{d}{Q}. \quad (150)$$

The last inequality follows by cross-multiplication and (149). The other odd blocks cost less than $1/10$ or $13/100$ and therefore fit the same strengthened bound. Positive lifting now proves (128). $\square$

The tail checker replays all scalar interval certificates and the finite remainder checks, using only Python's standard library and exact fractions. Run `optimized_tail_threshold_checker.py` with the `-S` flag. Its JSON output records the exact derivative lower bounds, endpoint margins, other retained-block bounds, coarse tail bound, and finite exhaustion count.

B.2 A direct quantitative coherent comparison

The following comparison avoids descendant truncation. Throughout this subsection $r \in \{3, 4\}$, $h = r - 2$, and the auxiliary space is $V_Q \otimes \text{Sym}^h U_Q$. Choose any fixed $a > 0$ and use the real exponent $M = aQ$ in the positive coherent kernel. Its positivity and the exact spectator marginal $(Q+1)/(M+1)$ do not require M to be an integer.

For a retained spin deficit z , let

$$Z_z = \frac{z(z-1)}{2}, \quad k_z = z + r - 1, \quad b_z = \frac{a(r/2)^2}{2(1-\theta_z)}, \quad c_z = 2k_z + b_z k_z (k_z + 1), \quad (151)$$

where $0 < \theta_z < 1$ is a proof parameter. Define

$$t_{3,z}(a) = (1 + 3a/2)^{-z-1}, \quad t_{4,z}(a) = (1 + a)^{-1}(1 + 2a)^{-z-1}. \quad (152)$$

These are the exact smallest eigenvalues of the limiting relative Gaussian kernels in degree z , including all multiplicities.

Lemma 5. *Suppose*

$$Q \geq z, \quad Q > \max(Z_z, c_z), \quad Q \geq \frac{2(1-\theta_z)}{a\theta_z^2}. \quad (153)$$

The spin-deficit- z multiplicity matrix of the coherent kernel obeys

$$A_{r,z,Q} \geq \left(1 + \frac{1}{2Q}\right) \left(1 + \frac{1}{Q}\right)^h \left(1 - \frac{Z_z}{Q}\right) \left(1 - \frac{c_z}{Q}\right) t_{r,z}(a) I. \quad (154)$$

Consequently its degree-resolved lifted comparison coefficient is

$$C_{r,z}(Q) = \frac{1}{a^h t_{r,z}(a) (1 + 1/(aQ))^h (1 + 1/(2Q)) (1 - Z_z/Q) (1 - c_z/Q)}. \quad (155)$$

For the three-body lift the corresponding form is $S_{4,Q} + H_Q$; for the four-body lift it is $2S_{4,Q} + H_Q$.

Proof. Use oscillator coordinates $x = (u, v_1, \dots, v_h) \in \mathbb{C}^{h+1}$, with the spectators symmetrized. Degree- z normalized occupation labels are identified with the spherical weight- z labels. The normalized center-of-mass annihilator is $a_\zeta = (\sqrt{2}a_u + \sum_s a_{v_s})/\sqrt{r}$. Let F_z be an isometric frame for its degree- z kernel. The exact diagonal matrix

$$D_{Q,z}(p; j_1, \dots, j_h) = \sqrt{\frac{(2Q)_p}{(2Q)^p}} \prod_{s=1}^h \sqrt{\frac{(Q)_{j_s}}{Q^{j_s}}}, \quad p + \sum_s j_s = z, \quad (156)$$

satisfies $J_+/\sqrt{Q} = D_{Q,z-1}^{-1}(\sqrt{r}a_\zeta)D_{Q,z}$. Thus $V_z = D_{Q,z}^{-1}F_z$ spans the exact spherical highest space. With $G_z = V_z^\dagger V_z$, its normalized frame is $F_{z,Q} = V_z G_z^{-1/2}$. The product inequality $\prod_i (1 - x_i) \geq 1 - \sum_i x_i$ gives

$$(1 - Z_z/Q)I \leq D_{Q,z}^2 \leq I, \quad I \leq G_z \leq (1 - Z_z/Q)^{-1}I. \quad (157)$$

Indeed, the sum of its diagonal product losses is $[p(p-1)/4 + \sum_s j_s(j_s-1)/2]/Q \leq Z_z/Q$.

In the stereographic coordinate α , the unit-mass spherical measure is $d^2\alpha/[\pi(1+|\alpha|^2)^2]$, and a spin of capacity L has coherent coefficients $(1+|\alpha|^2)^{-L/2}\sqrt{\binom{L}{i}}\alpha^i$. Rescale the pair coordinate to $u = \sqrt{2Q}\alpha$, and the spectators to $v_s = \sqrt{Q}\beta_s$. Set

$$X = |u|^2 + \sum_s |v_s|^2, \quad d_s = |v_s - u/\sqrt{2}|^2, \quad D_{\text{rel}} = \sum_s d_s. \quad (158)$$

The relative quadratic form satisfies $D_{\text{rel}} \leq \kappa X$ with $\kappa = r/2$: its nonzero eigenvalue is $3/2$ for $r = 3$, and its relative eigenvalues are $1, 2$ for $r = 4$.

For a multiplicity vector w , set $\psi = F_z G_z^{-1/2} w$ and define its coherent-overlap polynomial $P(x) = \sum_{\mathbf{n}} \bar{\psi}_{\mathbf{n}} x^{\mathbf{n}} / \sqrt{\mathbf{n}!}$. It is homogeneous of degree z ; the conjugation follows the bra-overlap

convention, and its Gaussian norm equals $\|\psi\|$. The factor $D_{Q,z}^{-1}$ in the spherical highest vector cancels the falling-factorial factors of the coherent coefficients exactly. Therefore

$$\begin{aligned} \langle w, A_{r,z,Q} w \rangle &= \mathcal{J}_Q \int_{\mathbb{C}^{h+1}} |P(x)|^2 W_Q(x) \prod_i \frac{d^2 x_i}{\pi}, \\ \mathcal{J}_Q &= (1 + 1/(2Q))(1 + 1/Q)^h, \\ W_Q(x) &= \prod_{i=0}^h \left(1 + \frac{|x_i|^2}{\ell_i Q}\right)^{-\ell_i Q - 2} \prod_{s=1}^h (1 - y_s)^{aQ}, \\ y_s &= \frac{d_s}{Q(1 + |u|^2/(2Q))(1 + |v_s|^2/Q)}, \quad \ell_0 = 2, \quad \ell_s = 1 \ (s \geq 1). \end{aligned} \quad (159)$$

Here $x_0 = u$ and $x_s = v_s$. The identity for y_s follows from $1 - s(\alpha, \beta) = |\alpha - \beta|^2 / [(1 + |\alpha|^2)(1 + |\beta|^2)]$, so $0 \leq y_s \leq 1$. The normalized spectator symmetrizer gives no additional factor in a highest-vector overlap.

The corresponding exact planar weight is $W_\infty = \exp(-X - aD_{\text{rel}})$. For $X \leq \theta_z Q/\kappa$, we have $y_s \leq d_s/Q \leq \theta_z$. The logarithmic-series bound $-\log(1 - y) \leq y + y^2/[2(1 - \theta_z)]$, together with $\log(1 + t) \leq t$ and $\sum_s d_s^2 \leq \kappa^2 X^2$, gives

$$W_Q \geq W_\infty \exp[-(2X + b_z X^2)/Q] \geq W_\infty [1 - (2X + b_z X^2)/Q]. \quad (160)$$

For $X > \theta_z Q/\kappa$, the bracket is nonpositive by (153), since $b_z(\theta_z Q/\kappa)^2/Q \geq 1$. Positivity of W_Q therefore makes the last bound in (160) valid on all of $\mathbb{C}^{h+1}$.

Normalize the positive measure $|P|^2 W_\infty \prod_i d^2 x_i / \pi$. At fixed angular direction ω , writing $x = \sqrt{X} \omega$ gives the radial density proportional to $X^{k_z-1} e^{-\lambda(\omega)X}$, with $\lambda(\omega) = 1 + aD_{\text{rel}}(\omega) \geq 1$. The exact Gamma-distribution moments therefore give $\mathbb{E}[X] \leq k_z$ and $\mathbb{E}[X^2] \leq k_z(k_z + 1)$, including after averaging over ω . Integrating the global weight bound yields

$$\int |P|^2 W_Q \geq (1 - c_z/Q) \int |P|^2 W_\infty. \quad (161)$$

The planar integral is bounded below by $t_{r,z}(a) \|G_z^{-1/2} w\|^2$. Equation (157) now proves (154). Inverting this bound, composing with $W_r, W_r^\dagger$, and applying the positive lift gives the claimed coefficient, because the exact marginal $\mu_Q = (Q + 1)/(aQ + 1)$ obeys

$$\frac{\mu_Q^h}{\mathcal{J}_Q} = \frac{a^{-h}}{(1 + 1/(aQ))^h (1 + 1/(2Q))}. \quad (162)$$

□

For uniform bounds at all $Q \geq Q_0$, define

$$\bar{C}_{r,z}(Q_0) = \max\{C_{r,z}(Q_0), a^{-h} t_{r,z}(a)^{-1}\}. \quad (163)$$

This bound applies even at small z where $C_{r,z}(Q)$ need not decrease. Indeed, the logarithm of $(1 + x/a)^h (1 + x/2)(1 - Z_z x)(1 - c_z x)$ is concave for $0 \leq x \leq 1/Q_0$, so its minimum is bounded by its endpoint minimum. If $Z_z + c_z > h/a + 1/2$, logarithmic differentiation gives a strictly negative derivative, and the simpler bound $C_{r,z}(Q) \leq C_{r,z}(Q_0)$ holds for every $Q \geq Q_0$.

In particular, a degree-resolved few-body error satisfying

$$E_{r,Q} \geq - \sum_{z=0}^D \delta_z H_{r,Q} \Pi_{z,Q}^{(r)}, \quad \delta_z \geq 0, \quad (164)$$

has lifted negative cost at most $\max_z \delta_z \bar{C}_{r,z}(Q_0)$ times the corresponding form. To see this, compare the single auxiliary weighted projector $\sum_z \delta_z \Pi_z$ with the same positive coherent kernel

in every spin block, and then lift once. The value of a must be common to all blocks of that kernel; θ_z may vary since it is only a proof parameter.

With $a = 1/12$ for $r = 3$ and $a = 1/16$ for $r = 4$, the following thresholds are obtained by exact rational evaluation of (155) and its monotonicity:

r	cutoff D	comparison constant	sufficient Q	θ
3	8	35	4505	1/14
3	8	36	1254	1/7
3	8	38	544	1/5
4	16	2100	4330	1/12
4	16	2200	2107	1/8

The threshold in each row is uniform over the retained degrees, since Z_z, c_z increase with z and $t_{r,z}(a)$ decreases. The constants 35, 2100 therefore hold simultaneously for $Q \geq 4505$, independently of particle number. The exact scalar checks are supplied in `optimized_coherent_check.py`; general rational parameters and degree-resolved bounds are implemented in `optimized_coherent_coefficients.py`.

B.3 A parameter-interval certificate for the finite head

We certify degree-dependent errors on the entire flux interval $Q \geq q_0$. The parameter is the real scalar $x = Q^{-1} \in [0, X]$, where $X = q_0^{-1}$; $x = 0$ denotes the planar limit. For each retained pair (r, z) and specified rational $\delta_{r,z} \geq 0$, the verifier proves

$$E_{r,Q}|_z + \delta_{r,z} H_{r,Q}|_z \geq 0 \quad \text{for every integer } Q \geq q_0. \quad (165)$$

Here restriction is to the highest-weight multiplicity space of deficit z ; rotational invariance extends the claim to the whole spin block. The interval enclosure establishes (165) throughout the real parameter interval.

We first express all finite forms in fixed polynomial coordinates. Use the canonical basis $|A]_{x,Q} = \prod_j (\sqrt{j!} a_j^\dagger)^{n_j} |0\rangle$ on the sphere, with A a partition of total weight M into n parts. The indexed creator x_j from the rational verifier remains distinct from the scalar $x = Q^{-1}$. The planar Gram entry is $g_A = \prod_j n_j! (j!)^{n_j}$. Define the scalar polynomials and their positive square roots by

$$P_j(x) = \prod_{\ell=0}^{j-1} (1 - \ell x), \quad T_p(x) = \prod_{\ell=0}^{p-1} (1 - \ell x/2), \quad d_j = \sqrt{P_j}, \quad e_p = \sqrt{T_p}, \quad d_A = \prod_{j \in A} d_j. \quad (166)$$

For $n \leq 4$ and $M \leq 16$, let $\mathcal{P}_{n,M}$ be the coordinate space of columns indexed by these complete fixed-weight occupation states. The invertible map to the spherical weight space is

$$\Phi_{n,M,Q} : \mathcal{P}_{n,M} \longrightarrow (\text{Sym}^n U_Q)_M, \quad c \longmapsto \sum_A c_A d_A^{-1} |A]_{x,Q}. \quad (167)$$

It is a polynomial-coordinate identification, not a unitary map. Its Gram matrix is

$$G_{n,M}(x) = \Phi_{n,M,Q}^\dagger \Phi_{n,M,Q} = \text{diag}(g_A/d_A^2). \quad (168)$$

All matrices with superscript `pol` below are pullbacks of physical quadratic forms by Φ , already including this metric.

Let $\hat{B}_p = \sum_{a=0}^p \partial_a \partial_{p-a}$, with ∂_j the canonical polynomial derivative. Direct substitution of the spherical pair coefficients gives the exact identity

$$B_{p,Q} \Phi_{n,M,Q} = \frac{\kappa_p}{e_p} \Phi_{n-2,M-p,Q} \hat{B}_p, \quad \kappa_p^2 = \frac{p!}{2^{p+1}}. \quad (169)$$

The polynomial highest-weight space is the fixed rational kernel of $L = \sum_{j \geq 1} j x_{j-1} \partial_j$. The exact raising identity is

$$Q^{-1/2} J_+ \Phi_{n,M,Q} = \Phi_{n,M-1,Q} L. \quad (170)$$

If U is a full rational basis of this kernel, then

$$\ker [U^\dagger H_{r,z}^{\text{pol}}(x) U] = \ker [U^\dagger H_{r,z}^{\text{pol}}(0) U] \quad (0 \leq x \leq X). \quad (171)$$

This follows by applying (169) to every pair annihilator; all scalar factors and Gram matrices are invertible. The pair-sandwich structure of the normal-ordered coefficients, together with invariance of $\ker H_r$ under rotations and spin projections, implies that the comparison form also annihilates this same kernel exactly.

The verifier chooses a fixed linear complement using principal pivot coordinates of the exact rational planar Hamiltonian form. Its principal matrix is positive definite by rational elimination. A further invertible rational upper triangular change of coordinates approximately normalizes this Hamiltonian form to the identity. Decimal arithmetic proposes its entries, which are rounded to exact rationals; the nonzero diagonal is then checked. Write the resulting canonical coordinate frame as U_a . Positivity on its image suffices, because the exact form annihilates the complementary kernel. A block of rank zero is identically zero and requires no positive-pivot test.

The interval verifier uses the following direct normal-order construction. In a fixed channel t , label row columns by 0 and $a = (p, j)$, put $i = p + j - t$, and define the exact rational covariance

$$C_{ab}^{(t)} = \sum_{\rho} y_{\rho,a} y_{\rho,b}, \quad h_a(x) = \frac{i!}{e_p d_i d_j}. \quad (172)$$

The sum is over the frozen rows in that channel. For $n = 3, 4$, define the integer row functional $\ell_{p;l_1, \dots, l_{n-2}}$ as the vacuum coefficient of $\partial_{l_1} \cdots \partial_{l_{n-2}} \widehat{B}_p$ on the canonical occupation basis at weight $p + \sum_s l_s$. Integer occupation multiplicities are included. Within this channel suppress the superscript t on C , and write $b = (p', j')$, $i' = p' + j' - t$. The channel's unaveraged three-body form is

$$\begin{aligned} & \sum_a C_{0a} \frac{h_a}{e_t} (\ell_{t;i}^\top \ell_{p;j} + \ell_{p;j}^\top \ell_{t;i}) \\ & + \sum_{a,b: i=i'} \frac{C_{ab} i!}{e_p e_{p'} d_j d_{j'}} \ell_{p;j}^\top \ell_{p';j'}. \end{aligned} \quad (173)$$

Its four-body form is

$$\sum_{a,b} C_{ab} h_a h_b \ell_{p;j,i'}^\top \ell_{p';j',i}. \quad (174)$$

Terms in (173) belong to weight $M = p + j$; terms in (174) belong to $M = p + j + i' = (p + j) + (p' + j') - t$. Summing channels therefore gives the full forms $K_{3,M}^{\text{pol}}$ and $K_{4,M}^{\text{pol}}$, supported at $M \leq 8$ and $M \leq 16$. To verify their coefficients, the physical functional $a_{l_1} \cdots a_{l_s} B_p$ after pullback is

$$\frac{\kappa_p \sqrt{l_1! \cdots l_s!}}{e_p d_{l_1} \cdots d_{l_s}} \ell_{p;l_1, \dots, l_s}. \quad (175)$$

Substitution of $\lambda = y_0/\kappa_t$ and $\alpha_{p,j} = y_{p,j} \sqrt{i! / (\kappa_p^2 j!)}$ into the exact normal-order identities cancels the square-root factorials and gives (173)–(174).

The remaining targets also have finite coordinate formulas:

$$\begin{aligned}
H_{n,M}^{\text{pol}}(x) &= \sum_{p=0}^M \frac{\kappa_p^2}{e_p^2} \widehat{B}_p^\top G_{n-2,M-p}(x) \widehat{B}_p, \\
S_{3,M}^{\text{pol}} &= H_{3,M}^{\text{pol}} G_{3,M}^{-1} H_{3,M}^{\text{pol}} - H_{3,M}^{\text{pol}}, \\
S_{4,M}^{\text{pol}} &= \sum_{p=0}^M \frac{\kappa_p^2 \kappa_{M-p}^2}{e_p^2 e_{M-p}^2} v_p^\top v_p, \quad v_p = \widehat{B}_{M-p} \widehat{B}_p.
\end{aligned} \tag{176}$$

Here v_p is a row functional to the vacuum; the intermediate diagonal maps cancel in the two successive pair annihilators. The four-body target is $(1 - c_9 - \mu)S_{4,M}^{\text{pol}}$.

In polynomial coordinates the rescaled spherical lowering operator is

$$R(x) = \sum_j (1 - jx)x_{j+1}\partial_j, \quad C_k(x) = R(x)^k U_a, \quad n_k(x) = \prod_{l=1}^k l[r - (2z + l - 1)x]. \tag{177}$$

Thus $\Phi C_k / \sqrt{n_k}$ is the normalized descendant of the chosen highest frame. With $D_3 = 8, D_4 = 16$, exact Schur averaging gives the highest-frame form

$$\overline{K}_{r,z}(x) = \frac{2+x}{r+(1-2z)x} \sum_{M=z}^{D_r} \frac{C_{M-z}(x)^\top \mathbf{K}_{r,M}^{\text{pol}}(x) C_{M-z}(x)}{n_{M-z}(x)}. \tag{178}$$

This formula applies to a nonorthonormal frame because the normalized ladder maps preserve its Gram matrix. The finite Hermitian matrix tested by the verifier is

$$F_{r,z}(x) = U_a^\top [\mathbf{T}_{r,z}^{\text{pol}}(x) + \delta_{r,z} H_{r,z}^{\text{pol}}(x)] U_a - \overline{K}_{r,z}(x), \tag{179}$$

where $\mathbf{T}_3 = \mathbf{S}_3$ and $\mathbf{T}_4 = (1 - c_9 - \mu)\mathbf{S}_4$.

Every scalar interval has integer endpoints divided by 2^{192} . Products and reciprocals round lower endpoints down and upper endpoints up using integer floor and ceiling division. Square roots use integer square roots of the appropriately scaled endpoints, with an increment of the upper endpoint unless its square is exact. All denominator intervals must exclude zero and all square-root intervals must be nonnegative; otherwise verification stops.

A Taylor interval record of order m encloses the exact representation

$$f(x) = P_f(x) + x^{m+1} \rho_f(x), \quad P_f(x) = \sum_{k=0}^m c_k x^k, \quad 0 \leq x \leq X. \tag{180}$$

It stores intervals for every fixed coefficient c_k and for all values of ρ_f on the full interval. The certificates here use $m = 3$. Addition is componentwise. If $\text{tail}_{>m}$ discards degrees at most m from a polynomial, the product remainder is exactly

$$\rho_{fg} = \frac{\text{tail}_{>m}(P_f P_g)}{x^{m+1}} + P_f \rho_g + P_g \rho_f + x^{m+1} \rho_f \rho_g. \tag{181}$$

The polynomial quotient in the first term is well defined also at zero. Interval Horner evaluation of this identity encloses the product remainder on the whole interval.

For an inverse, its Taylor coefficients satisfy

$$b_0 = c_0^{-1}, \quad b_k = -c_0^{-1} \sum_{i=1}^k c_i b_{k-i}, \quad \rho_{1/f} = -\frac{\rho_f P_b}{f}, \quad P_b = \sum_{k=0}^m b_k x^k. \tag{182}$$

The last identity holds because the polynomial part of fP_b is exactly one. For the positive square root, the corresponding exact formulas are

$$b_0 = \sqrt{c_0}, \quad b_k = \frac{c_k - \sum_{i=1}^{k-1} b_i b_{k-i}}{2b_0}, \quad \rho_{\sqrt{f}} = \frac{\rho_f - \rho_{P_b^2}}{\sqrt{f} + P_b}. \quad (183)$$

The remainder identity rationalizes $\sqrt{f} - P_b$. These identities refer to the exact coefficients; rounding intervals may enclose a nonzero width around a canceled coefficient without affecting the algebraic cancellation. Structural induction using (181)–(183) proves that the final record encloses every matrix entry throughout $[0, X]$.

After symmetric averaging of the matrix records, write the exact form as

$$F_{r,z}(x) = \sum_{i=0}^m A_i x^i + x^{m+1} \mathcal{E}(x). \quad (184)$$

All coefficient and remainder matrices are symmetric. If $\mathcal{E}_{ij}(x) \in [\ell_{ij}, u_{ij}]$, a uniform remainder bound is

$$\gamma = X^{m+1} \max_i \sum_j \max\{|\ell_{ij}|, |u_{ij}|\}, \quad \|x^{m+1} \mathcal{E}(x)\| \leq \gamma. \quad (185)$$

The computed γ is rounded upward. This is the maximum absolute row-sum bound for a symmetric matrix in the chosen finite coordinate space.

For $s = x/X \in [0, 1]$, define the Bernstein control matrices

$$B_k = \sum_{i=0}^k \frac{\binom{k}{i}}{\binom{m}{i}} X^i A_i, \quad \sum_{i=0}^m A_i x^i = \sum_{k=0}^m \binom{m}{k} s^k (1-s)^{m-k} B_k. \quad (186)$$

The scalar weights are nonnegative and sum to one. The verifier proves $B_k - \gamma I > 0$ by interval $LDL^\top$ elimination, requiring each pivot interval to have strictly positive lower endpoint. Hence every actual symmetric matrix enclosed by that interval matrix is positive definite. If a control matrix is too conservative, repeated arithmetic averaging performs de Casteljau subdivision into the two half intervals. The verifier requires both halves to pass, recursively, using the same valid γ .

Equations (185) and (186) prove $F_{r,z}(x) \geq 0$ for the whole real interval. The common-kernel identity (171) extends this to the full highest space; invertible coordinate pullback and rotational invariance give (165).

The standard-library script `optimized_head_interval.py` takes `--qmin`, an exact rational shift JSON through `--shifts`, and an output path through `--result`. Its saved records include the rational frames, every Taylor coefficient and remainder interval, all Bernstein/LDL pivot intervals, and the exact shifts for the nonzero blocks. A complete run checks all $9 + 17 = 26$ relative blocks.

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